Semiconductors & Devices

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Semiconductors

Introduction

On the basis of concentration of charge carriers or electrical conductivity in solids, we already have studied conductors and insulators. We now introduce another kind of solid known as semiconductor. These solids conduct electricity in the presence of an applied electric field, but the conductivity is very low compared to conventional metal conductors. Silicon is the example of a semiconductor, its electrical conductivity is about 10^{11} times lower than copper and about 10^{13} times higher than molten quartz (insulator). Another distinguishing feature of a semiconductor is that its electrical conductivity increases with increasing temperature.

Type Of Solid	Conductivity (mho/m)
Metals	$10^{2} - 10^{8}$
Semiconductors	$10^{-6} - 10^{5}$
Insulators	$10^{-19} - 10^{-11}$

In general, semiconductors are of two types –

(a) Elemental semiconductors : Si and Ge

(b) Compound semiconductors : Examples are :

Inorganic : CdS, GaAs, CdSe, InP, etc.

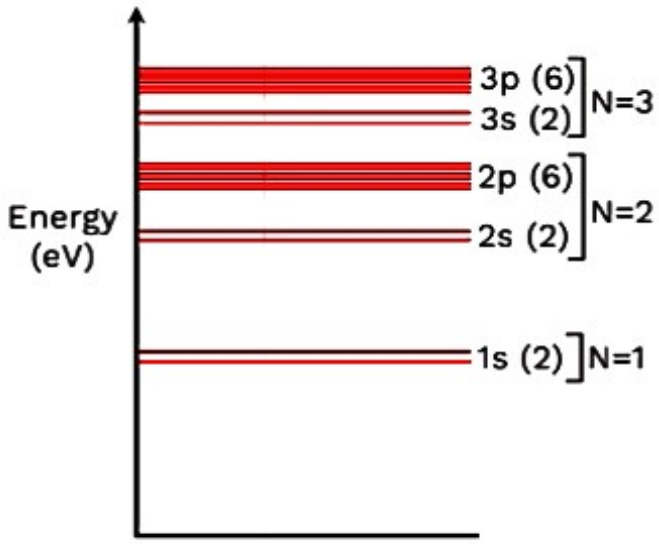
Organic : anthracene, doped pthalocyanines, etc.

Organic polymers : polypyrrole, polyaniline, polythiophene, etc.

It is necessary to understand the mechanism of conduction in solids, let us discuss qualitatively, formation of energy bands in solids (specially silicon).

The Energy Band Theory of Crystals

A crystal is basically a group of atoms or molecules. These atoms or molecules have fundamental structural unit. A crystal is series of these identical units and it is regularly repeated in three dimensions. The energy states of a crystal is different from an isolated atom.



Energy levels are further subdivided in sub-energy levels (i.e. s, p, d, f). Electronic configuration of Si atom is $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^2$. It is represented by energy level diagram in figure below. In case of single isolated atom, the electrons in any orbit possess definite energy. However, an atom in a crystal is very much influenced by closely packed neighbouring atoms. As a result, the electron in any orbit of such an atom can have a range of energies rather than a single energy.

An electron possesses energies in the range known as "Energy band".

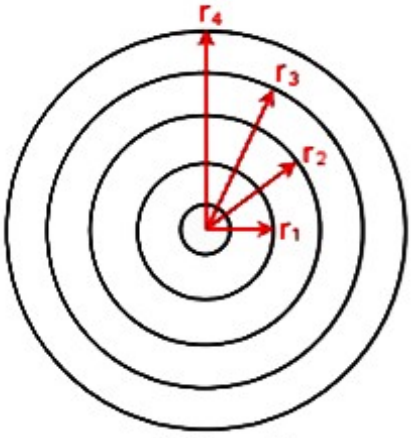


Figure 1

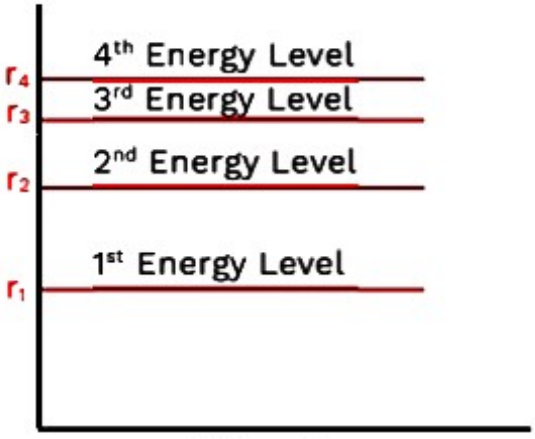


Figure 2

Figure 1, shows the orbits of different electrons around nucleus for an isolated atom.
Figure 2, is showing the energy level of an isolated atom, where each orbit is having single energy level.

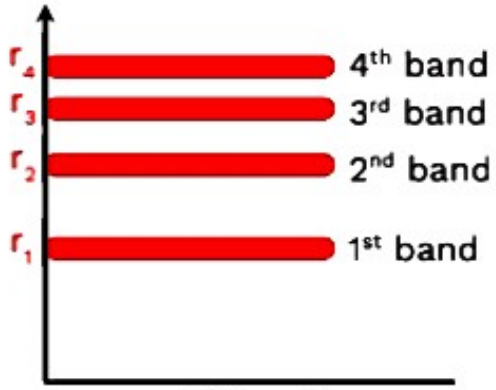


Figure 3

For a crystal, the electron in each orbit has a range of energies. According to Pauli exclusion principle, no two electrons can have same energy, therefore, electrons in first orbit have slightly different energies. A crystal contains millions of first orbit electrons, the slightly different energy levels form a band and known as 1st energy band as shown in figure 3. The electrons spinning in the in the first orbit can have any energy that lie in this band. Similarly, electrons of second orbit form second energy level band and so on.

Let us have an example of Si crystal to understand the "Energy band Theory". Consider that the Si crystal is having N no. of atoms. We know that outermost shell (n = 3) is having two sub-shells (s and p). These two sub-shells can have a maximum 8 number of electrons. Therefore, there will be 8 energy levels in the outermost shell of Si-atom. Since only 4 electrons lie in outermost shell, so out of 8, only 4 energy levels (or energy states) are filled, remaining 4 energy levels are empty.

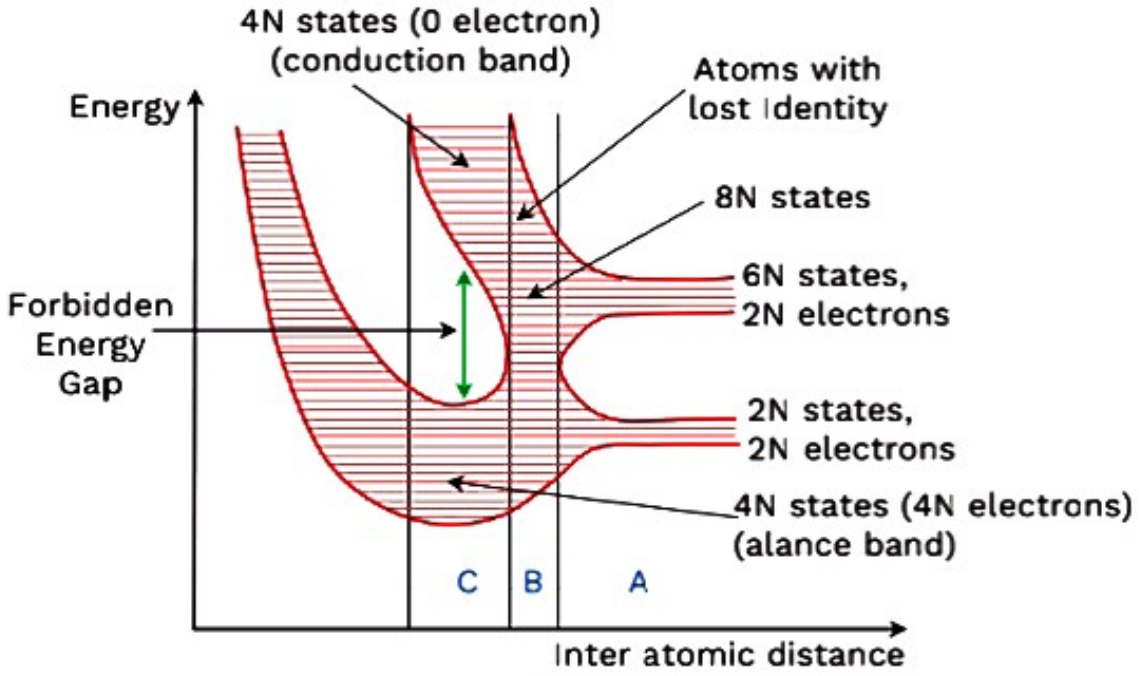
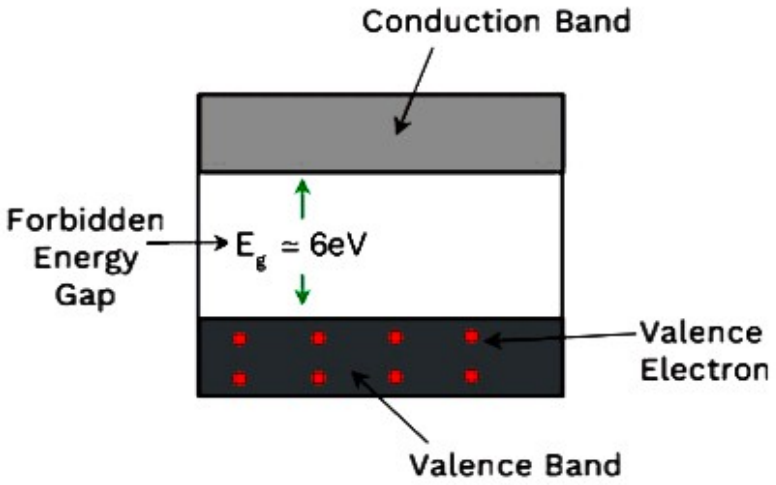


Figure 4

If distance between neighbouring N atoms is large, all the atoms behave as if they are isolated and have identical pattern. These atoms will be having 8N energy states, out of which 2N states will be in 3s sub-shell and other 6N states will be in 3p sub-shell and 3s and 3p sub-shells are separated by an energy gap. It is shown in figure 4 (portion A).
As the distance between atoms decreases, interaction between atoms increases. Because of close coupling between atoms, the atoms losses their own identity and become an electronic system which must obey Pauli Exclusion Principle.
At a specific distance 6N energy states of 3p subshell and 2N energy states of 3s subshell merged with each other and there will be total 8N energy states without any significant energy gap between them (figure 4, portion B).
If spacing between atoms is further decreased, the total energy levels start getting divided into two parts (4N, 4N) and an energy gap is established between them (figure 4, portion C). This energy gap is called Forbidden Gap.
At 0 K temperature, no electron can have energy corresponding to forbidden gap. These discrete energy levels are represented by horizontal lines in energy diagram having negligible separation among them and, therefore, look like a band. Lower energy band is called Valence band and upper energy band is called Conduction band. These bands are separated by forbidden gap.
Since every semiconductor has only four electrons in its outer shell, therefore, it will be having structure same as Si crystal.

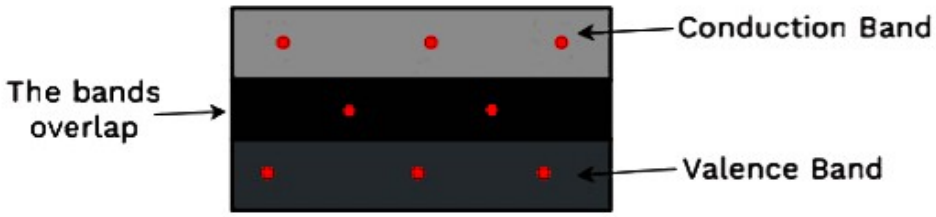
Insulator



Insulators are poor conductor of electricity. They have completely filled valence band and empty conduction band. These two bands are being separated by a forbidden gap of around 6eV. Normally an electron cannot acquire such a high amount of energy so it cannot jump from valence band to conduction band and hence conductivity is poor.

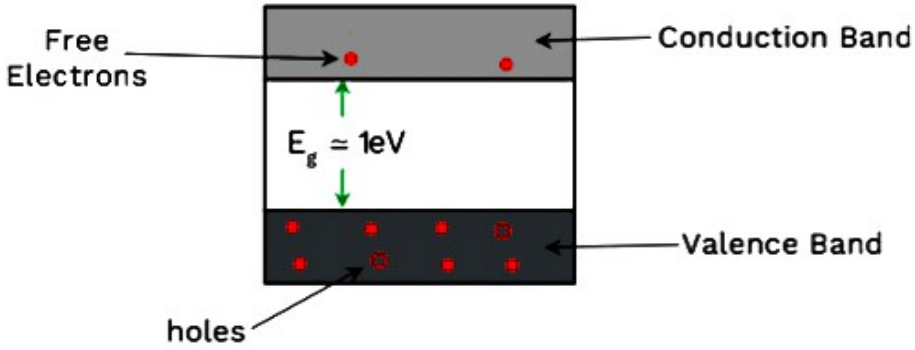
However, as the temperature increases, some of electrons of valence band acquire sufficient energy to cross over to the conduction band. Therefore, the resistance of insulator decreases with the increase in temperature. Hence, an insulator has negative temperature coefficient of resistance. Diamond and Glass are examples of insulator.

Metal



In a metal, conduction band overlap with valence band, so forbidden gap is not present and electrons present in valence band can easily move into conduction band. Therefore, large number of charge carriers are present in metal that constitute high conductivity of metals.

Semiconductor



Semiconductor is a substance whose conductivity lies between metal and insulator. Forbidden energy gap is nearly 1eV. At 0K, semiconductors behave like insulators, since no electron can have any thermal energy to jump into conduction band. At room temperature some valence electrons acquire sufficient energy to cross forbidden gap and become free/conduction electrons and hence conductivity increases with temperature. Therefore, semiconductor have negative temperature coefficient. Once they jump into conduction band, they leave vacancies in valence band. These vacancies are known as holes.

Comparison of different substances			
S. No.	**Substance**	**Nature**	**Conductivity (mho/m)**
1.	Glass	Insulator	1.11×10^{-7}
2.	Germanium	Semiconductor	1.67
3.	Copper	Conductor	9×10^{7}

Hole

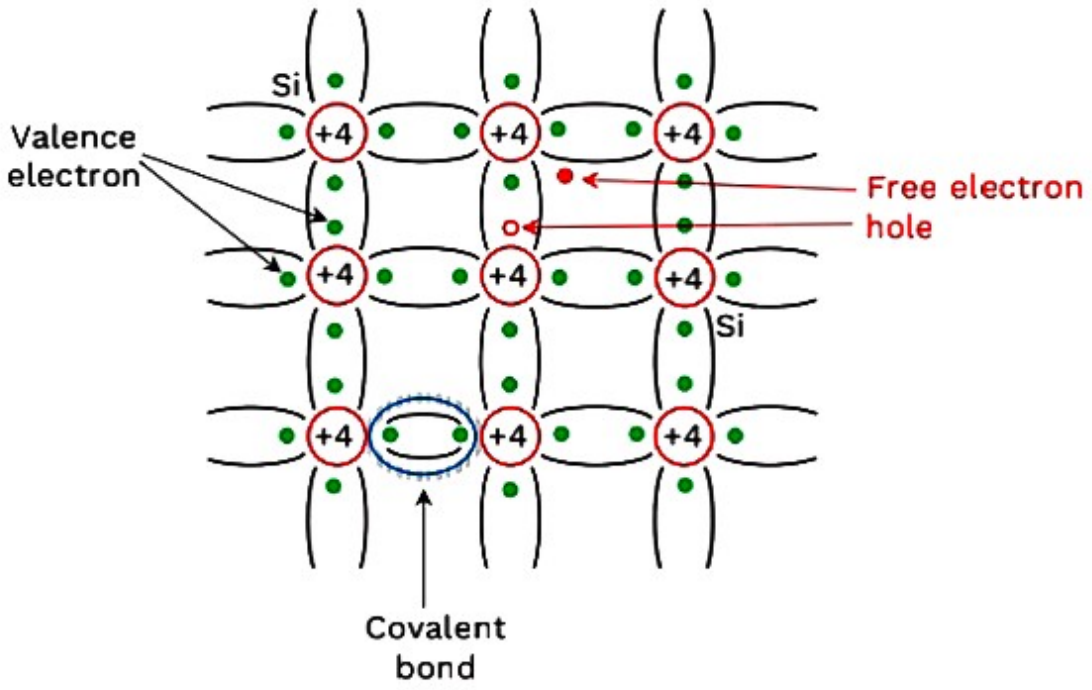
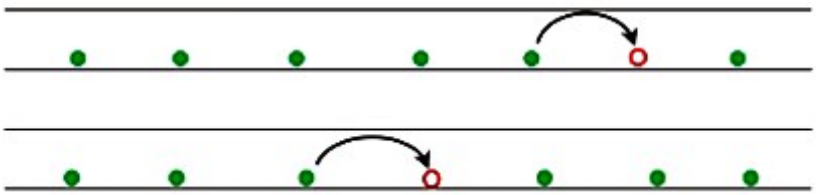


Figure shows a Si crystal with broken covalent bond. Each atom has four electrons in outer shell, each electron is shared by neighbouring atom and hence covalent bond is formed. At room temperature, some covalent bonds are broken. Energy required to break the bond is 1.1eV for Si and 0.72 eV for Ge at room temperature. The absence of electron in the covalent bond or an incomplete covalent bond is called a hole. This hole acts as a positive charge. For one electron set free, one hole is created. Therefore, due to thermal energy, pairs of electron and hole are created, there being as many holes as the free electrons.

Current Carrying Mechanism of Hole



Assuming that a hole exists in a covalent bond. Electron from neighbouring atom fills this hole and leaves a hole at its own position. This process will continue. Valence electrons move towards right and it appears as holes are moving towards left.
Basically current carried by valence electrons in valence band is said to be the hole current (i_h) and current carried by free electrons (or conduction electrons) in conduction band is termed as electron current (i_e). The total current flow in Si crystal will be –

$$i = i_h + i_e$$

Types of Semiconductors

(a) Intrinsic (pure) Semiconductors :- These are the substances without any impurity. Their conductivity is lower than extrinsic semiconductors. At high temperature, some of the covalent bonds are broken and few electron-hole pairs are created.
These free electrons and holes are carriers of electricity in an intrinsic semiconductor.

(b) P-type Extrinsic Semiconductor :- They are impure semiconductors with P-type (trivalent) of impurities doped in. They have majority of holes as charge carriers.

(c) N-type Extrinsic Semiconductor :- When N-type (pentavalent) impurity is added to intrinsic semiconductor they become N-type extrinsic semiconductors. Majority carriers are electrons in N-type semiconductors.

Generation and Recombination of Charges

In an intrinsic semiconductor, due to thermal energy or photoelectric radiation, new electron-hole pairs are created, while at the same time some other electron-hole pairs disappear due to recombination. Recombination is a process in which a conduction electron falls into empty covalent bond (hole). Before recombination, the electron-hole pair exist for some time. This time is called mean life time. This time vary in the range of 10^{-4} sec to 10^{-9} sec. Life time of hole and electron is indicated by τ_h and τ_e respectively.

Intrinsic Concentration

In a pure (intrinsic) semiconductor, the concentration of electrons (n_e) is equal to concentration of holes (n_h).

$\Rightarrow$ $n_e = n_h = n_i$

where n_i is called intrinsic concentration

n_i depends on temperature by following relationship.

$$n_i^2 = AT^3 e^{-\left[\frac{(E_g)_0}{KT}\right]}$$

where A = Constant that is independent of T
T = Absolute temperature
$(E_g)_0$ = Forbidden energy gap at 0 K
K = Boltzmann's constant = 1.38×10^{-23} J/K

	Ge	Si
n_i at room Temp. (300 K)	2.5×10^{13} / cm^3	1.5×10^{10} / cm^3

Forbidden Energy Gap

The separation between conduction band and valence band on the energy level diagram is called Forbidden energy gap (E_g). For a semiconductor E_g depends upon the temperature by following relationship.

For Ge : $E_g(T) = [0.785 - 2.23 \times 10^{-4}T]$ eV where, T is temp. in K.

For Si : $E_g(T) = [1.21 - 3.6 \times 10^{-4}T]$ eV

	E_g at 0 K	E_g at 300 K
Ge	0.785 eV	0.72 eV
Si	1.21 eV	1.1 eV

Q. **At which temperature, the forbidden energy gap for Ge and Si become equal?**

Sol. $(E_g)_{Si} = (E_g)_{Ge}$

$1.21 - 3.6 \times 10^{-4} T = 0.785 - 2.23 \times 10^{-4} T$

$1.37 \times 10^{-4} T = 0.425$

$T = 3102.2$K

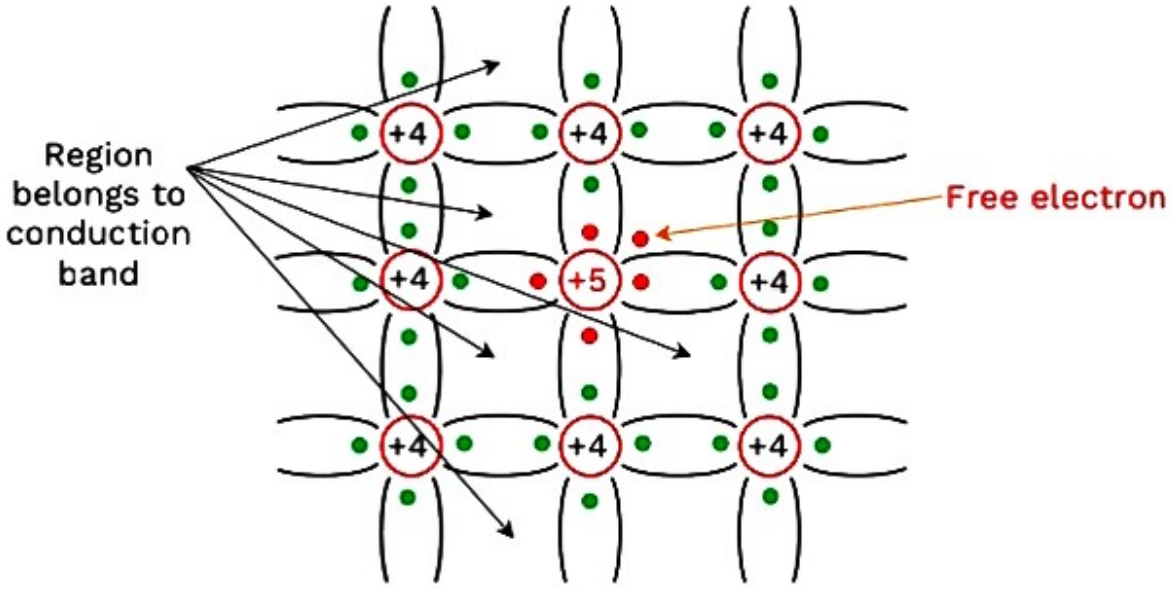
Types of Doping

If trivalent or pentavalent impurities are added to intrinsic semiconductor, extrinsic (impure) semiconductor is formed.

Trivalent / Acceptor Impurities → **B, Al, Ga, In, Tℓ**

Pentavalent / Donor Impurities → **N, P, As, Sb, Bi**

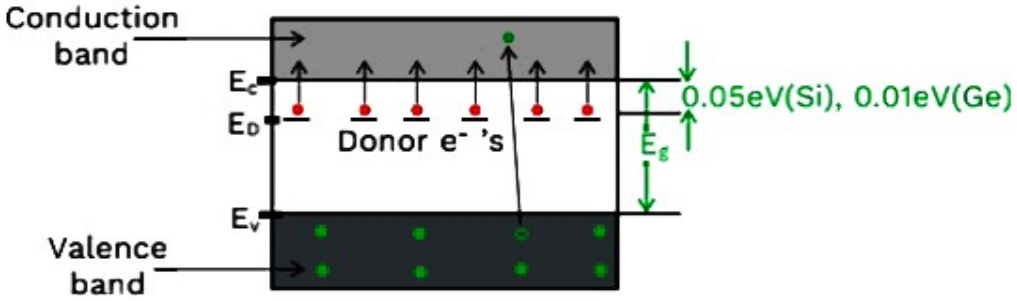
N-type Semiconductor



Pentavalent/Donor atoms have five valence electrons, four electrons form covalent bond with atoms of intrinsic semiconductor and fifth is loosely bound. This requires low amount of energy that is very easily available at room temperature, therefore it aces like free electron and carries the current.

Almost every Donor atom provides one conduction electron at room temperature, the electrons (negative charge) become the majority charge carries and therefore this type of semiconductor is known as N-type extrinsic Semiconductor. Since every pentavalent atom donate one electron, it is also called Donor impurity.

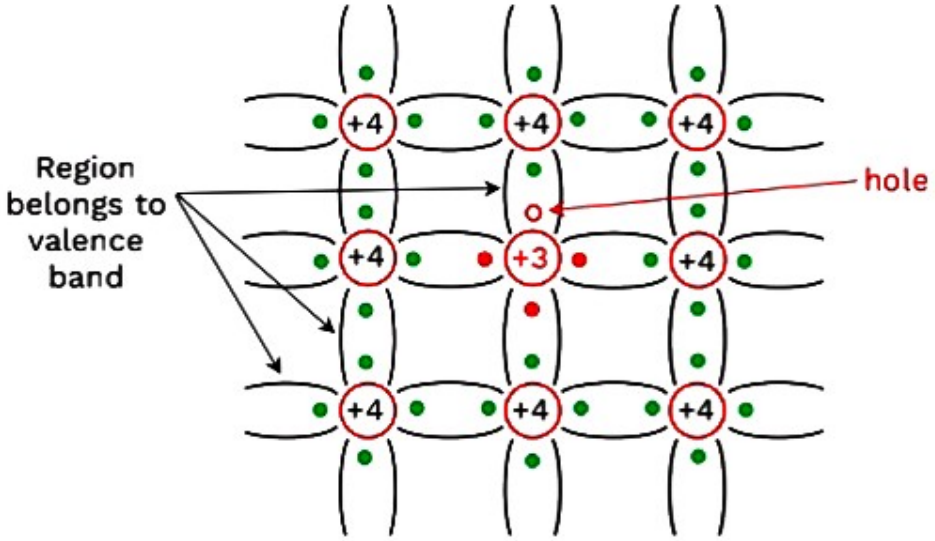
Every donor atom acquires positive charge and becomes positive ion, when electron leaves it. Since weight of an ion is much more than an electron, therefore, it is almost immobile and termed as immobile positive ion.



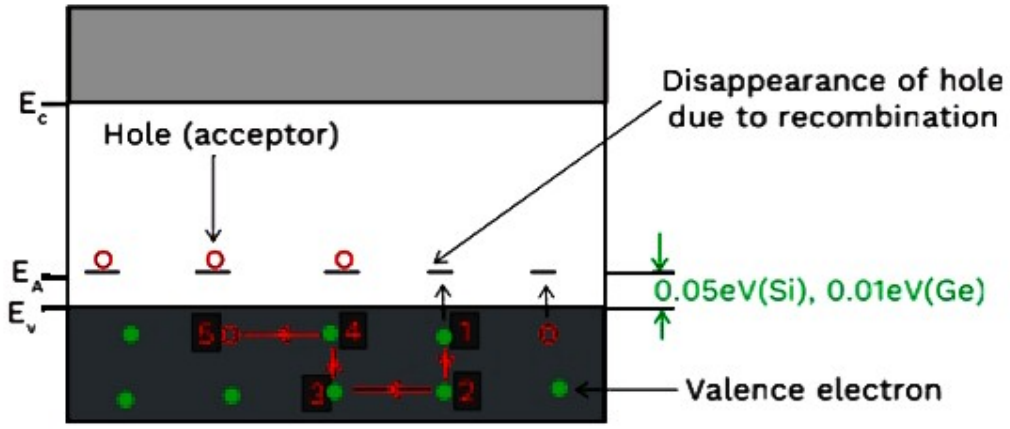
Effect of donor impurities on energy band structure

Addition of donor impurity creates new donor energy level (E_D) in forbidden energy gap, it lies just below conduction band. Energy gap is 0.05eV for Si and 0.01eV for Ge. This low energy level is available at room temperature, so all donor electrons jump from donor level to conduction level quite easily and large number of free electrons are available, thus increasing the conductivity.

P-type Semiconductor



Trivalent atom has only three electrons in outer shell, so they have vacancy of one electron (known as hole) to complete the covalent bond. Therefore, every trivalent atom provides one hole. Total number of holes in an extrinsic semiconductor depends upon number of added trivalent atoms. Since every trivalent atom accepts one electron so these type of impurities are known as acceptor impurities. Current conduction in P-type semiconductor is due to holes that have positive charge, therefore acceptor impurities are called P-type impurities and semiconductor so formed is called P-type extrinsic semiconductor.



Effect of acceptor impurities on energy band structure

Doping of acceptor impurities creates new energy levels called acceptor energy level (E_A) in forbidden energy gap. It lies just above valence band. These new energy levels contain holes. Electrons from valence band move into acceptor level (E_A) and recombine with holes, so they leave holes in valence band.

Hole is created at position 1 due to movement of electron from position 1 to acceptor level. Electron from position 2, jump to position 1 and create hole at position 2. This shifting of position continues and movement of hole takes place from position 1 to position 5 as shown in above figure. Therefore, in P-type semiconductors, the current conduction in valence band is due to valence electrons. Since it appears as holes are moving in valence band, so it is termed as hole current.

Every acceptor atom acquires negative charge after accepting electron and is called immobile negative ion.

Effect of Doping on Properties of Semiconductor

The doping has following effects on Semiconductor :

1. Increased concentration of charge carriers (holes or electrons).
2. Conductivity (σ) increases (as $\sigma \propto$ concentration of charge carriers).
3. It creates extra energy level, for N-type new energy level is created near conduction band and for P-type near valence band.
4. Concentration of electrons in conduction band increases for N-type semiconductor.
5. Concentration of holes in valence band increases for P-type semiconductor.
6. For N-type semiconductor, concentration of holes (minority carriers) falls below the level of holes in intrinsic semiconductor. Since N-type semiconductor have large number of electrons, therefore, rate of recombination increases and concentration of holes decreases. Although concentration of electrons also decreases due to recombination but percentage change is negligible, considering large concentration of electrons in N-type semiconductor.

7. Concentration of free electrons fall below intrinsic level for P-type semiconductor in the same manner as explained above.
8. Doping make the current mainly dependent upon only one type of majority charge carrier (either hole or electron).
9. For N-type semiconductor, concentration of electrons much exceed the holes, therefore, electrons are called majority carriers and holes as minority carriers. Similarly, the holes are said to be majority carriers and electrons are called as minority carriers in P-type semiconductor.

Mass Action Law

Under thermal equilibrium, the product of concentration of free electrons (n_e) and concentration of holes (n_h) is a constant, independent of amount of donor and acceptor doping level. This is called mass action law. Mathematically, we have-

$$n_e.n_h = n_i^2$$

where n_i = intrinsic concentration.

1. **For N-type Semiconductor :**

   Suppose N_D is the concentration of donor atoms in intrinsic semiconductor. At room temperature almost every donor atom donates one electron and therefore concentration of free electrons due to donor atoms is N_D. As we know,

   $$n_e = N_D \quad \text{and} \quad n_e >> n_h$$

   According to mass action law,

   $$n_e n_h = n_i^2 \Rightarrow N_D n_h = n_i^2$$

   Therefore, concentration of minority charge carrier or holes,

   $$n_h = \frac{n_i^2}{N_D}$$

2. **For P-type Semiconductor :**

   Suppose N_A is the concentration of acceptor atoms in intrinsic semiconductor. At room temperature almost every acceptor atom donate one hole and therefore concentration of holes due to acceptor atoms is N_A.

   $$n_h = N_A \quad \text{and} \quad n_h >> n_e$$

   According to mass action law,

   $$n_e n_h = n_i^2 \Rightarrow N_A n_e = n_i^2$$

Therefore, concentration of minority charge carrier or electrons,

$$n_e = \frac{n_i^2}{N_A}$$

Mobility

Drift velocity (v) $\propto$ applied electric field (E)

$$v = \mu E \Rightarrow \mu = \frac{v}{E} \Rightarrow \mu = \frac{e\tau}{m}$$

The proportionality constant is called mobility (μ) and has a unit of $cm^2/V\text{-}sec$. Also τ is mean free time and m is mass of electron (9.1×10^{-31} kg).

	Ge	Si
Mobility of electron (μ_e)	3800 $cm^2/V\text{-}cm$	1300 $cm^2/V\text{-}cm$
Mobility of hole (μ_h)	1800 $cm^2/V\text{-}cm$	500 $cm^2/V\text{-}cm$

Mobility of conduction electrons is more than valence electrons (holes) since conduction electrons are more easily affected from electric field than valence electrons. Valence band is inner band of atoms where influence of applied electric field is least.

1. **Dependence of Mobility over Temperature :**

   $$\mu \propto T^{-n} \text{ where } n = \frac{3}{2}$$

   At T = 0 K, μ = 0 for all charge carriers.

2. **Dependence of Mobility over Electric Field Intensity :**

   μ = constant, for $E < 10^3$ V/cm

   $\mu \propto \frac{1}{\sqrt{E}}$, for $10^3 < E < 10^4$ V / cm

   $\mu \propto \sqrt{E}$, for $E < 10^4$ V / cm

Conductivity

Conductivity depends upon number of charge carriers and mobility. It is expressed as-

$$\sigma = \frac{n_e^2 \tau}{m} \Rightarrow \sigma = ne\mu$$

1. **For Intrinsic Semiconductor :**

   Intrinsic semiconductor is bipolar in nature (i.e. it has both holes and electrons as charge carriers). So conductivity can be expressed as-

   $$\sigma = n_e e\mu_e + n_h e\mu_h$$

$\because \quad n_e = n_h = n_i$

$\therefore \quad \sigma = n_i e(\mu_e + \mu_h)$

where μ_e and μ_h are the mobilities of conduction electrons and valence electrons (holes) respectively.

2. For N-type Semiconductor :

$\because \quad n_e >> n_h \quad$ and $\quad n_e \simeq N_D$

$\therefore \quad \sigma \simeq N_D e\mu_e$

3. For P-type Semiconductor :

$\because \quad n_h >> n_e \quad$ and $\quad n_h = N_A$

$\therefore \quad \sigma \simeq N_A e\mu_h$

Q. **An intrinsic sample of germanium crystal has a hole density of 10^{13} per cm^3 at the room temperature. When doped with antimony so as to contain 10^{15} impurity atoms/cm^3, the hole density decreased to 10^{11} per cm^3 at the same temperature. Calculate the majority carrier density.**

Sol. For intrinsic semiconductor $n_e = n_h = n_i \Rightarrow n_i = n_h = 10^{13} / cm^3$

After doping concentration of hole $n_h = 10^{11}/cm^3$

$\because \quad n_e n_h = n_i^2$

$$\therefore \quad n_e = \frac{n_i^2}{n_h} = \frac{\left(10^{13}\right)^2}{10^{11}} \quad \Rightarrow \quad n_e = 10^{15} / cm^3$$

Q. **Find the resistivity of intrinsic silicon. Intrinsic carrier concentration (n_i) = $1.5 \times 10^{16} / m^3$; Hole mobility (μ_h) = 0.048 m^2 / V–sec; Electron mobility (μ_e) = 0.135 m^2/V–sec.**

Sol. Intrinsic Semiconductor :

Given, $\quad n_i = n_e = n_h = 1.5 \times 10^{16}/m^3$

$\mu_e = 0.135\ m^2/V–sec$

$\mu_h = 0.048\ m^2/V–sec$

Conductivity, $\quad \sigma = n_e e\mu_e + n_h e\mu_h$

$= n_i e\ (\mu_e + \mu_h)$

$= 1.5 \times 10^{16} \times 1.6 \times 10^{-19} \times 0.183$

$= 4.392 \times 10^{-4}$ mho/m

Resistivity, $\rho = \frac{1}{\sigma}$

$\rho = 2277\,\Omega - m$

Q. **When intrinsic silicon is doped with a pentavalent impurity of one impurity atom for each 60 million Si atoms, calculate the resistivity of the doped Si. No. of Si atoms (N) = 4.5×10^{28} / m^3; Electron mobility (μ_e) = 0.135 m^2/V–sec.**

Sol. Doped Semiconductor :

$\because \quad n_e \simeq N_D = \frac{1}{60 \times 10^6} \times 4.5 \times 10^{28}$

$n_e \simeq N_D = 7.5 \times 10^{20}\ / m^3$

Resistivity, $\rho = \frac{1}{\sigma} \simeq \frac{1}{N_D e \mu_e}$

$\rho = \frac{1}{7.5 \times 10^{20} \times 1.6 \times 10^{-19} \times 0.135}$

$\rho = 0.0617\,\Omega - m$

Q. **Find the conductivity of Si with acceptor impurity of 1 part in 5×10^7. Given at 300 K, μ_h = 500 cm^2/V–s ; No. of Si atoms = 5×10^{22} / cm^3.**

Sol. P–type Semiconductor :

$n_h \simeq N_A = \frac{1}{5 \times 10^7} \times 5 \times 10^{22}$

$N_A = 10^{15}\ / cm^3$

Conductivity,

$\sigma \simeq N_A e \mu_h$

$\sigma = 10^{15} \times 1.6 \times 10^{-19} \times 500$

$\sigma = 0.08$ Mho/cm

Q. **For a Germanium single crystal of N–type, the mobility of electrons in 3600 $cm^2/V–sec$ and conductivity is 0.6 mho/cm. Calculate the value of the carrier density neglecting the contribution due to holes.**

Sol. In N–type Semiconductor:

$$n = n_e + n_h \simeq n_e \simeq N_D$$

$$\because \quad \sigma \simeq N_D e\mu_e \simeq ne\mu_e$$

$$\therefore \quad n = \frac{\sigma}{e\mu_e} = \frac{0.6}{1.6 \times 10^{-19} \times 3600}$$

$$n = 1.04 \times 10^{15} / cm^3$$

Q. **The resistivity of doped material is 0.039 ohm–cm. Calculate electron density and hole density. Given: $n_i = 2.5 \times 10^{13}/cm^2$, $\mu_e = 3600\ cm^2/V–sec$, $\mu_h = 1700\ cm^2/V–sec$.**

Sol. $$\because \quad n_e n_h = n_i^2 = (2.5 \times 10^{13})^2$$

$$\therefore \quad n_e = \frac{6.25 \times 10^{26}}{n_h} \qquad \ldots(1)$$

$$\because \quad \sigma = \frac{1}{\rho} = n_e e\mu_e + n_h e\mu_h$$

$$\therefore \quad n_e(3600) + n_h(1700) = \frac{1}{0.039 \times 1.6 \times 10^{-19}}$$

$$36 n_e + 17 n_h = 1.603 \times 10^{18}$$

$$36\left(\frac{6.25 \times 10^{26}}{n_h}\right) + 17 n_h = 1.603 \times 10^{18}$$

$$17 n_h^2 - 1.603 \times 10^{18} n_h + 2.25 \times 10^{28} = 0$$

$$n_h = \frac{1.603 \times 10^{18} \pm \sqrt{(1.603 \times 10^{18})^2 - 4(17)(2.25) \times 10^{28}}}{2(17)}$$

By taking +ve' sign-

$$n_h = 9.43 \times 10^{16} \text{ holes} / cm^3$$

and $n_e = \frac{n_i^2}{n_h} = \frac{(2.5 \times 10^{13})^2}{9.43 \times 10^{16}}$

$\Rightarrow \quad n_e = 6.63 \times 10^9$ electrons / cm^3

By taking '–ve' sign, we can get another possible answer – (using binomial approximation)

$$n_h = \frac{1.603 \times 10^{18} - 1.603 \times 10^{18}\left[1 - 2.977 \times 10^{-7}\right]}{2(17)}$$

$n_h = 1.403 \times 10^{10}$ holes / cm^3

and $n_e = \frac{n_i^2}{n_h} = \frac{6.25 \times 10^{26}}{1.403 \times 10^{10}}$

$n_e = 4.45 \times 10^{16}$ electrons / cm^3

Q. **A uniform silver wire has resistivity = 1.54 × 10^{-8} Ω–m room temperature. Electric field E = 1.2 V/m is applied along the wire. Find average drift velocity and mobility of electrons. Electron density = 6 × 10^{28}/m^3.**

Sol. $\because \quad \sigma = ne\mu$

$\therefore \quad \mu = \frac{1}{\rho ne} = \frac{1}{1.54 \times 10^{-18} \times 6 \times 10^{28} \times 1.6 \times 10^{-19}}$

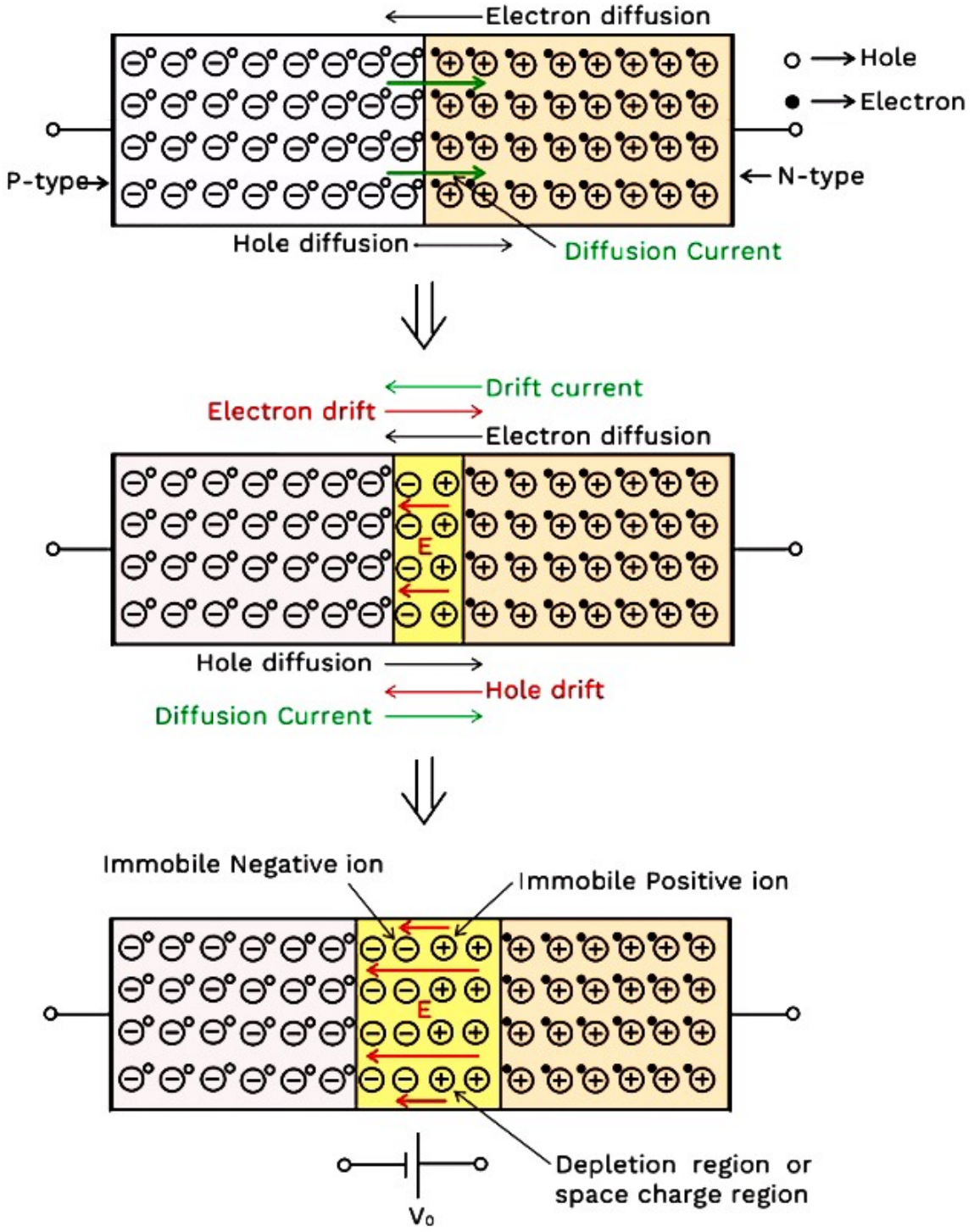
$\mu = 6.76 \times 10^{-3}\ m^2 / V - \text{sec}$

Also, drift speed $v_d = \mu E \quad \Rightarrow \quad v_d = 6.76 \times 10^{-3} \times 1.2$

$v_d = 8.12 \times 10^{-3}$ m / s

P-N Junction

A P-N junction is the basic building block of many semiconductor devices like diode, transistor, etc. A clear understanding of the junction behaviour is important to analyse the working of other semiconductor devices.



Two important processes occur in the formation of the PN junction: one is diffusion and other is drift. We know that, the concentration of electrons in an N-type semiconductor is more than the concentration of holes. Likewise, in a P-type semiconductor, the concentration of holes is greater than the concentration of electrons. During the formation of the PN junction and due to the concentration gradients on the P and N sides, holes diffuse from the P side to the N side (P → N) and electrons diffuse from the N side to the P side (N → P). This charge transport motion gives rise to diffusion currents across the junction.

When an electron diffuses out of the N→P, it leaves behind an N-side ionised donor. The donor which is ionised (positive charge) is immobile because it is bonded to the surrounding atoms. As the electrons continue to diffuse away from the N → P, a positively charged layer on the N side of the junction forms.

Likewise, when a hole diffuses from P→N due to a concentration gradient, it leaves an immobile ionised acceptor(negative charge). As the holes continue to diffuse, a negatively charged layer (or negatively charged region) on the P-side of the junction is formed. The spatially charged region on either side of the junction is called the depletion region because the electrons and holes involved in the initial motion across the junction deplete the region of its majority carriers. The thickness of the depletion zone is one tenth of a micrometer. In this region, an electric field (E) directed from positive to negative charges develops. Due to this field, an electron from the P-side of the junction moves towards N-side, and a hole from the N-side of the junction moves towards P side. The movement of charge carriers due to the electric field is called drift. Thus, a drift current, opposite to the diffusion current, begins.

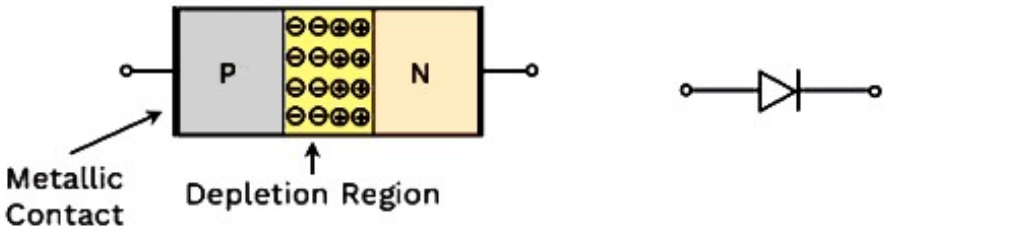
Initially, the diffusion current is large and the drift current is small. As diffusion continues, the spatially charged regions on either side of the junction expand, thereby increasing the strength of the electric field and thus the drift current. This process continued upto the moment till the diffusion current equals the drift current. Thus, a P-N junction is formed. In a P-N equilibrium junction, there is no net effective current.

The loss of electrons from the N region and the gain of electrons by the P region cause a potential difference at the junction of the two regions. Since this potential tries to prevent the movement of electrons from the N region to the P region, it is often called the barrier potential (V_0) or the contact potential or the cut-in potential.

Q. **Can we take a P-type semiconductor slab and physically connect it to another N-type semiconductor to get a P-N junction?**

Sol. No we cannot. Any plate, no matter how flat, will have a roughness much greater than the distance between the crystals (–2 to 3Å) and therefore cannot be in continuous atomic-level contact. The junction will act as a discontinuity for moving charge carriers.

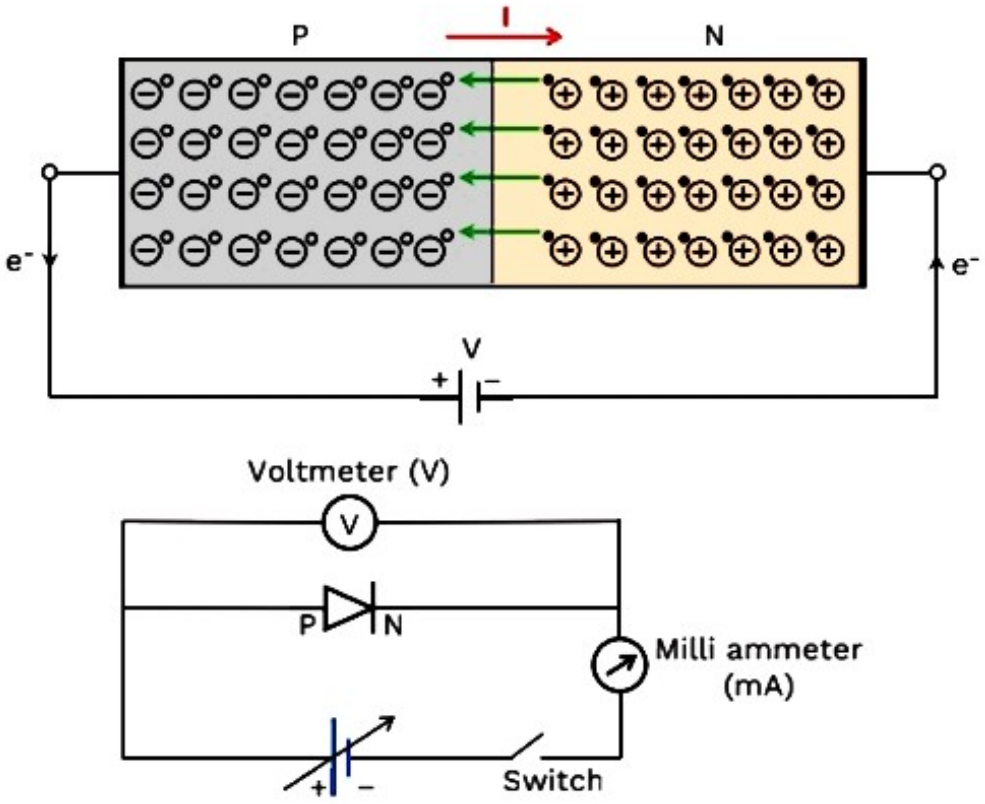
P-N Junction Diode



P–N junction diode **Symbol of P–N junction diode**

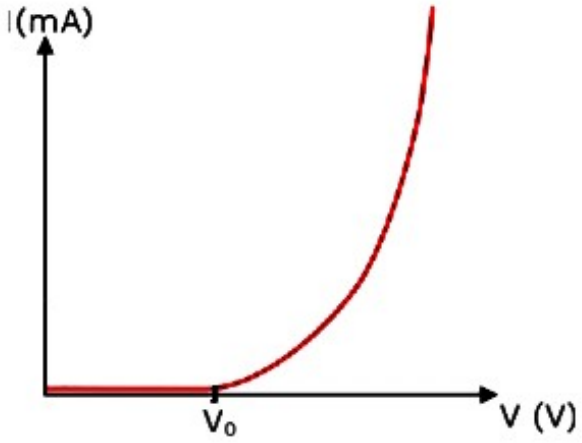
A P-N junction diode is basically a two terminal device with metallic contacts provided at the ends for the application of an external voltage.
The direction of the arrow shows the original direction of current (under forward bias). The equilibrium barrier potential can be changed by applying an external voltage V across the diodes.

1. P-N Junction Diode Under Forward Bias :



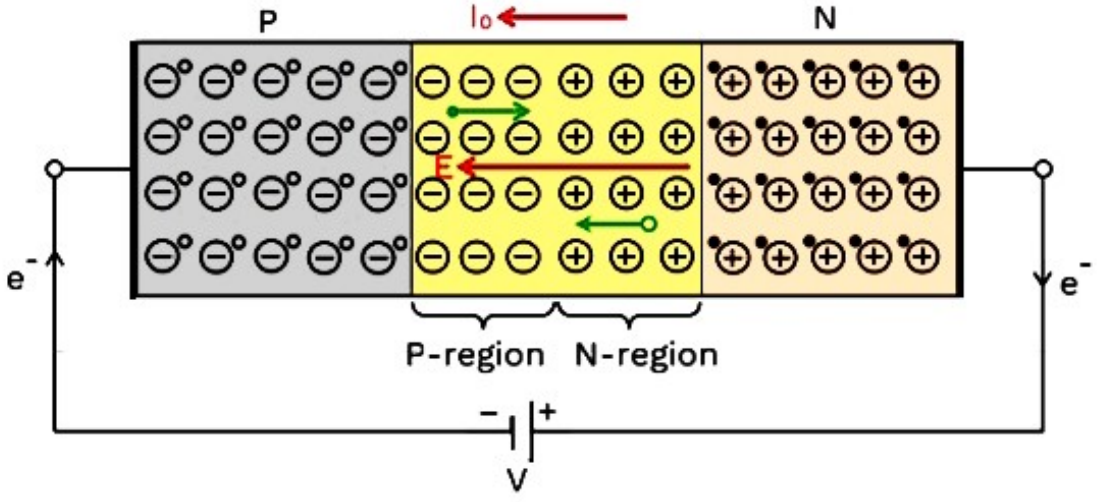
When positive terminal of external biasing battery is connected to P-side of diode and negative terminal to the N-side of diode, the diode or P-N junction is said to be the forward biased.

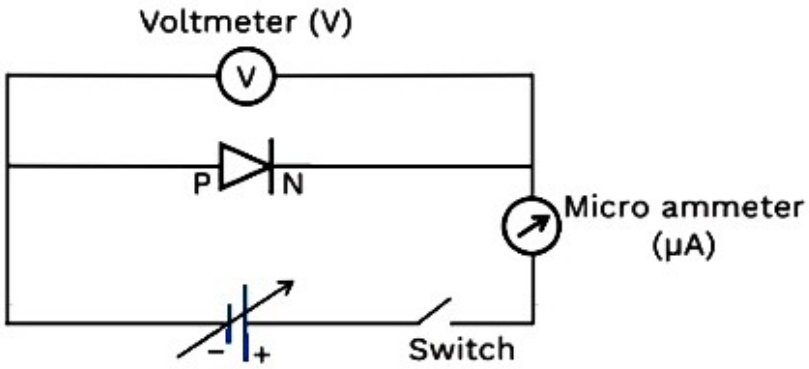
Negative terminal of battery repels electrons of N-side towards the junction, therefore they move into depletion region and merge with the positive ions and reconvert positive ions into neutral donor atoms. Similarly, holes are repelled by positive terminal of battery and they merge with negative ions and neutralize them. With decrease in concentration of positive and negative ions, depletion width decreases and finally it completely disappears when applied forward voltage is equal to cut-in potential.
Now majority carriers can cross the junction and they constitute a large current which is called forward current. It is directed from P to N-region. The junction offers low resistance in the forward biased condition.



In forward bias, the current is almost negligible, till the applied voltage crosses the cut-in voltage (~0.2V for Ge diode and ~0.7V for Si diode). After this voltage, the diode current increases significantly (exponentially), even for a very small increase in the diode bias voltage.

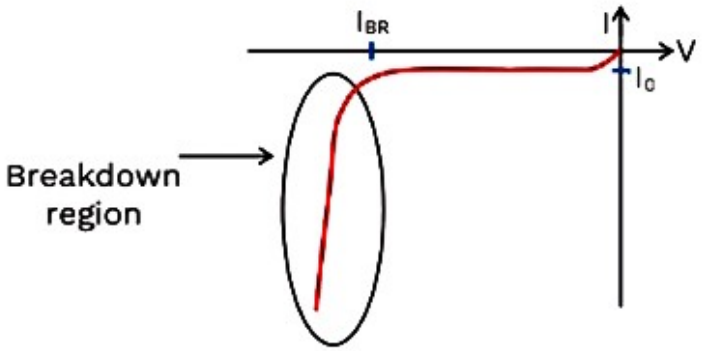
2. P-N Junction Diode Under Reverse Bias:





Reverse biasing a P-N junction means connecting negative terminal of external DC battery to P-side and positive terminal to N-side of diode. Negative terminal of battery attracts holes from P-side, therefore, holes move away from the junction. As holes leave acceptor atoms, these atoms acquire negative charge and become negative immobile ions. Similarly, positive terminal of battery attracts electrons from N-side and due to movement of electrons away from the junction, more donor atoms are uncovered and become positive ions. Since positive and negative ions increase in number, therefore, thickness of depletion layer and contact potential increases. The greater the reverse bias, the wider the depletion layer becomes. The depletion layer stops growing when contact potential becomes equal to potential of biasing battery.

Since barrier potential increases with increase in reverse bias, therefore, majority carriers (holes in P-side and electrons in N-side) couldn't cross the junction. As a result, the current due to majority carriers will be zero. The P-N junction offers very high resistance under reverse bias.

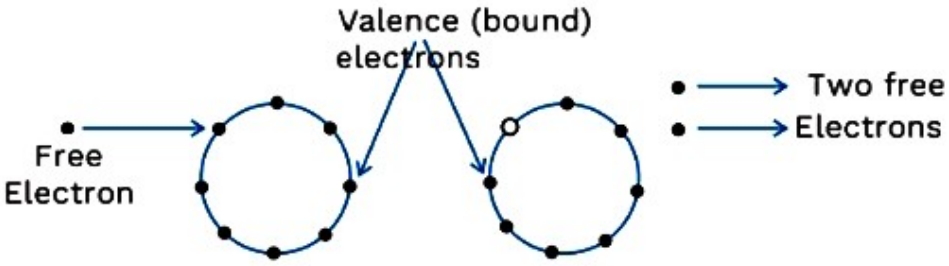


Minority Current (I_0) :

Holes of N-region and electrons of P-region are called minority carriers. Electric field in the depletion region helps the minority carriers in crossing the junction as shown in previous topic. For minority carriers, reverse biasing the P-N junction acts like forward bias. Current constituted by minority carriers under reverse bias condition is called minority current (I_0). It's direction is opposite to forward current. It depends only upon temperature and is

independent of applied reverse voltage. For a fixed temperature, minority current (I_0) reaches its maximum value quickly and does not change with increase in reverse bias potential, it is also called as reverse saturation current. It is of the order microamperes for Ge diode and nanoamperes for Si diode. This is one of the reason why silicon dominates the field of semiconductor devices.

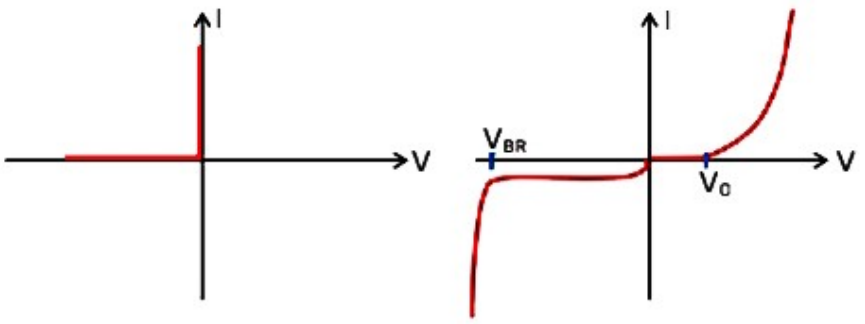
Avalanche Breakdown :



When reverse voltage increases continuously, then there will a stage where reverse current increases suddenly to a very high value and diode may be damaged due to heating. This is known as breakdown of diode.

Because of reverse voltage, the thermally generated free electrons of P-side (minority carriers) pushed towards N-side. As electrons move through depletion layer, they acquire energy from electric field present inside depletion layer and accelerate. These high speed free electrons strike the atoms and dislodge the valence electrons. We can see a case in above figure, a free electron strikes an atom and imports its energy to valence electron and dislodge it. As a result, two free electrons are available. These two electrons gain energy and further dislodge two valence electrons. It is like a chain reaction in atom bomb and an all-out avalanche occurs and diode current increases tremendously. The voltage at which junction breaks down is called reverse break over voltage (V_{BR}).

V-I Characteristics :



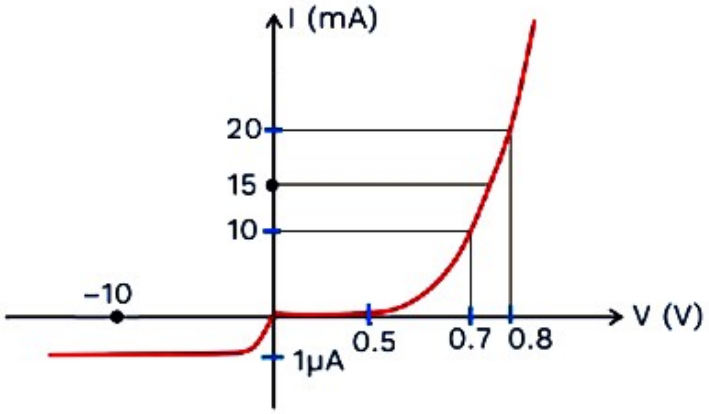
V-I curve for an ideal diode **V-I curve for a practical diode**

An ideal diode offers zero resistance in forward bias (behave as a short circuit) and infinite resistance in reverse direction (behave as open circuit).
For diodes, we define a quantity called dynamic resistance (r_d) as the ratio of small change in voltage ΔV to a small change in current ΔI :

$$r_d = \frac{\Delta V}{\Delta I}$$

In reverse bias, diode resistance is variable, it increases as applied reverse voltage increases.

Q. **The V-I characteristics of a silicon diode is shown in figure. Calculate the resistance of the diode at**
(a) I_D = 15 mA and **(b) V_D = –10 V.**



Sol. (a) ∵ Curve is almost straight between I = 10 mA to I = 20 mA

∴ Dynamic resistance (when I = 15 mA) is-

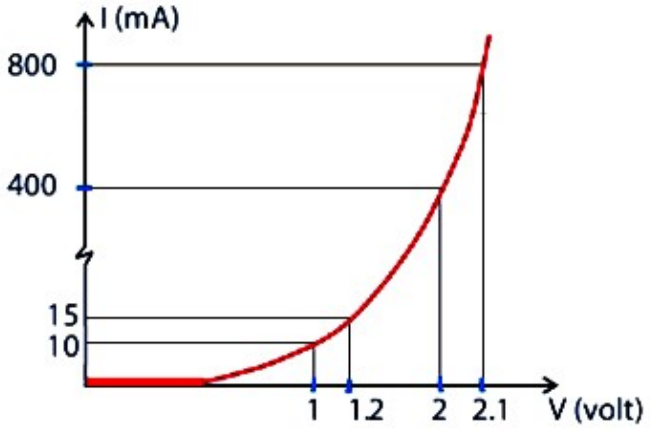
$$r_d = \frac{\Delta V}{\Delta I} = \frac{0.1\,V}{10\,mA} = 10\,\Omega$$

(b) V = –10 V and I = –1 μA

Therefore, static resistance is-

$$R = \frac{V}{I} = \frac{-10V}{-1\mu A} = 10^7\,\Omega$$

Q. **The I-V characteristic of a P-N junction diode is shown in figure. Find the approximate dynamic resistance of the P-N junction when (a) a forward bias of 1 volt is applied, (b) a forward bias of 2 volt is applied.**



Sol. (a) ∵ Curve is almost linear between 1V to 1.2V,
∴ Dynamic resistance in this region is-

$$R = \frac{\Delta V}{\Delta I} = \frac{0.2\,V}{5\,mA} = 40\,\Omega$$

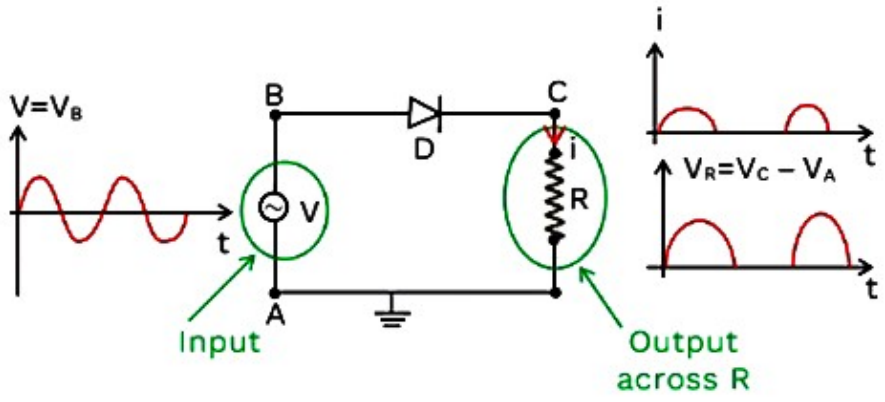
(b) ∵ Curve is almost linear between 2V to 2.1V,
∴ Dynamic resistance in this region is-

$$R = \frac{\Delta V}{\Delta I} = \frac{0.1V}{400\,mA} = 0.25\,\Omega$$

P-N Junction as a Rectifier

A rectifier is a device which converts an AC into a DC.

1. Half-wave Rectifier :



We can use a P-N junction as a rectifier because it permits current in one direction only. Above figure shows an AC source connected to a load resistance through a diode. The potential at B varies with time as $V = V_0 \sin(\omega t + \phi)$. During the positive half cycle, $V > 0$ and $V_B > V_A$. In this case, the junction is forward-biased and a current i is established in the load resistance R in the direction C to A.
During this half cycle,

$$i = \frac{V_0 \sin(\omega t + \phi)}{R + R_{junction}}$$

And the potential difference across it is-

$$V_R = \frac{RV_0 \sin(\omega t + \phi)}{R + R_{junction}}$$

where $R_{junction}$ = resistance offered by diode.

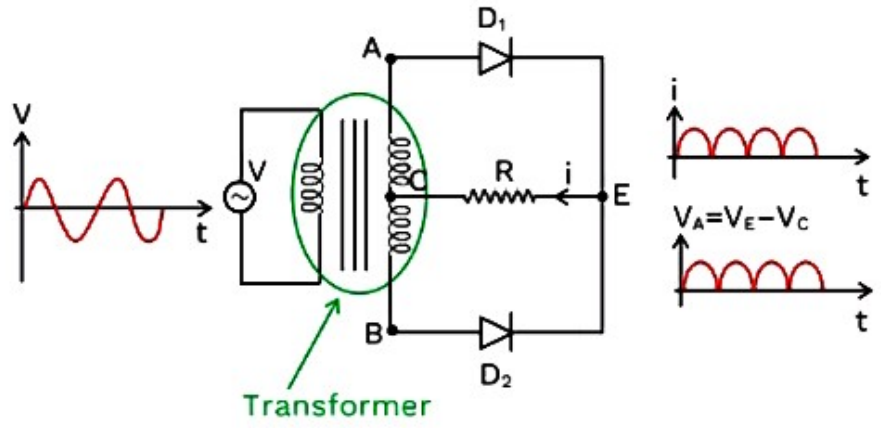
In forward bias, $R_{junction} << R$ so that

$$i \approx \frac{V_0}{R} \sin(\omega t + \phi)$$

During the next half cycle, $V < 0$ and the potential at the point B becomes smaller than that of A. The junction is thus reverse-biased and offers a large resistance during this half cycle and there is only a negligible current in the circuit. The current in the resistance is thus unidirectional. This is called half-wave rectification because there is practically no current during alternate half cycles.

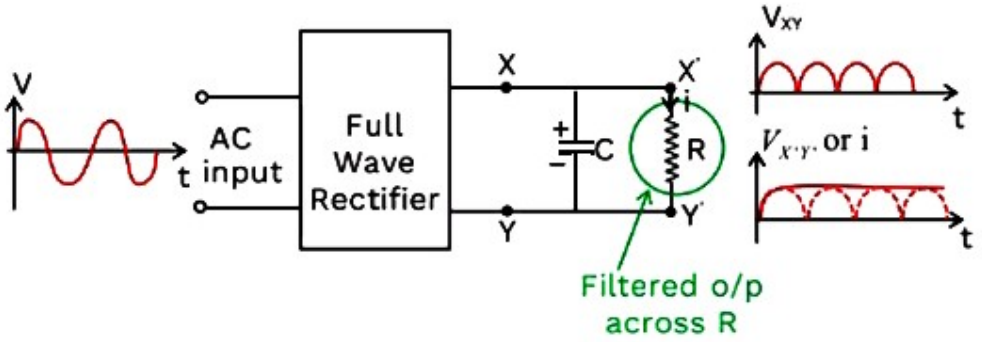
2. Full-wave Rectifier :

A full wave rectification can be achieved by using two diodes as shown in figure below.



The AC potential difference is obtained across the secondary coil of a transformer and is connected in the circuit. In one-half cycle, $V_A > V_C > V_B$ so that the junction D_1 conducts but D_2 does not. The current is from A to D_1 to E to C. In the next half cycle, $V_B > V_C > V_A$ so that D_2 conducts whereas D_1 does not. The current is from B to D_2 to E to C. In both the half-cycles, the current in the load resistance is from E to C.

Filters



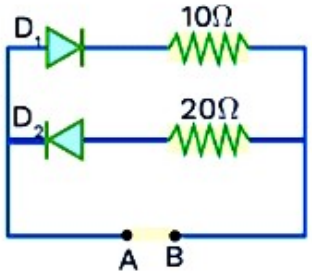
The rectifier voltage has a pulse in the form of a half-sinusoidal shape. Although it is one-way, it has no fixed value. To obtain a stable DC output from the pulse voltage, a capacitor is usually connected to the output terminals (in parallel with the load R). We can also use an inductor in series with R for the same purpose. Since these additional-circuits seem to filter the AC ripple and supply pure DC voltage, they are called filters.

As the voltage across the capacitor increases, it charges. If there is no external load, it is still charged at the peak voltage of the rectified output. When there is a load, it discharges on the load and the voltage across it starts to drop. During the next half cycle of the rectified output, it is again charged to the peak value. The rate of the fall of the voltage across the capacitor in proportional to $\frac{1}{RC}$ or inverse of time constant.

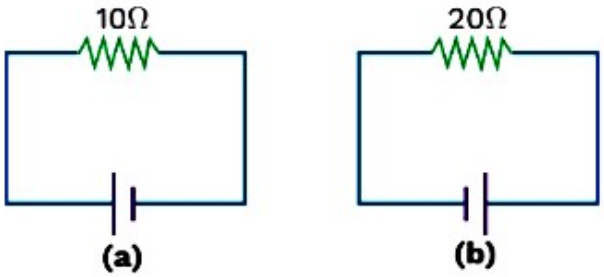
A good filter uses large value of C. The output voltage of C.

The output voltage obtained using a capacitor input filter is closer to the peak voltage of the rectified voltage. Such type of filter is most widely used in power supplies.

Q. **A 2V battery may be connected across the points A and B as shown in figure. Assume that the resistance of each diode is zero in forward bias and infinity in reverse bias. Find the current supplied by the battery if the positive terminal of the battery is connected to (a) the point A (b) the point B.**



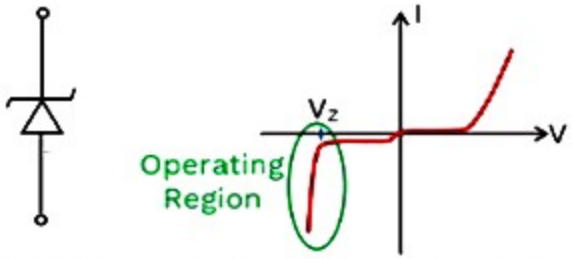
Sol. (a) When the positive terminal of the battery is connected to the point A, the diode D_1 is forward-biased and D_2 is reverse-biased. The resistance of the diode D_1 is zero, and it can be replaced by a resistance-less wire. Similarly, the resistance of the diode D_2 is infinity, and it can be replaced by a broken wire. The equivalent circuit is shown in figure (a). The current supplied by the battery is $2V/10\Omega = 0.2$ A.



(b) When the positive terminal of the battery is connected to the point B, the diode D_2 is forward-biased and D_1 is reverse biased. The equivalent circuit is shown in figure (b). The current through the battery is 2 V/20 Ω = 0.1 A.

Zener diode

It is a special purpose semiconductor diode, designed to operate under reverse bias in the breakdown region and used as a voltage regulator.



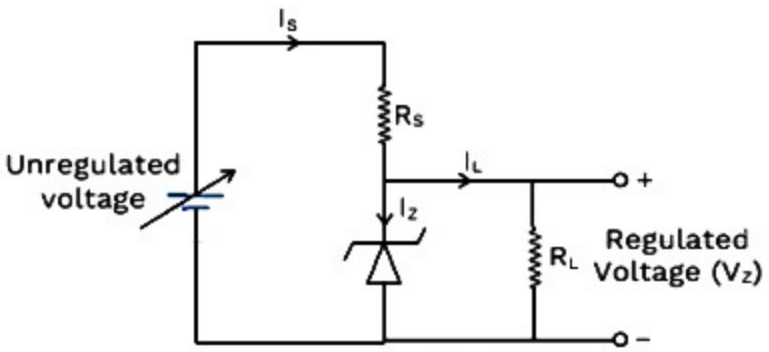
Symbol **I-V characteristics of Zener diode**

Zener diode is fabricated by heavily doping both P- and N-sides of the junction. Due to this, depletion region formed is very thin (< 10^{-6}m) and the electric field of the junction is extremely high (~5 × 10^6 V/m) even for a small reverse bias voltage of about 5 V. When the applied reverse bias voltage V reaches the breakdown voltage V_z, there is a large change in current. Its Not like that a small change in the reverse bias voltage can cause a big variation in the current following the breakdown voltage V_z. In other words, even though the current through the Zener diode changes widely, the Zener voltage remains constant.

Zener Breakdown :

The movement of electrons (minority carriers) from P→N and holes from N→P is known to be the cause of reverse current. The electric field at the junction becomes considerable as the reverse bias voltage is increased. When the reverse bias voltage $V=V_z$ is set, the electric field strength is high enough to remove valence electrons from the P-side host atoms and accelerate them to the N-side. The strong current detected at the breakdown is due to these electrons. Internal field emission, also known as field ionisation, is the emission of electrons from host atoms as a result of a strong electric field. Field ionisation requires an electric field on the order of 10^6 V/m. This is referred to as a Zener breakdown.

Zener Diode as a Voltage Regulator :



Zener diode as a DC voltage regulator

When the AC input voltage of a rectifier varies, so does the rectified output. A Zener diode is used to obtain a steady DC voltage from a rectifier's DC unregulated output.
The Zener diode is reverse biased by connecting the unregulated DC voltage (filtered output of a rectifier) to it through a series resistance R_s. As the input voltage rises, the current flowing through the R_s and Zener diode rises as well. This raises the voltage drop across R_s while leaving the voltage across the Zener diode same. This is because, even while the current through the Zener diode changes in the breakdown zone, the Zener voltage remains constant.
Similarly, as the input voltage drops, so does the current through the R_s and Zener diode. Thus without any change in the voltage across the Zener diode, the voltage drop across R_s diminishes. As a result, each change in the input voltage causes a change in the voltage drop across R_s without a change in the voltage across the Zener diode. Hence we can say that as a voltage regulator, the Zener diode is used. The Zener diode should be chosen based on the needed output voltage and the series resistance R_s.
Also, for a good load regulation, I_z should be much larger than I_z (in general, at least 5 times).

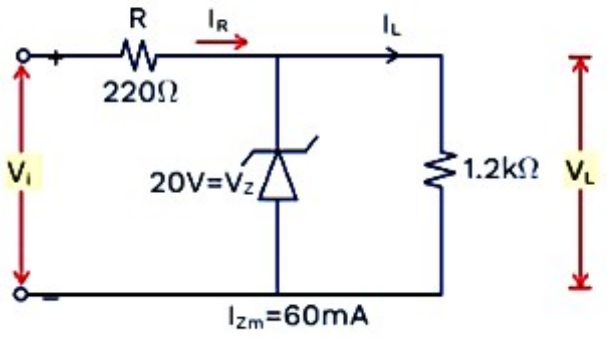
Q. **In a Zener regulated power supply a Zener diode with V_z = 6.0V is used for regulation. The load current is to be 4.0 mA and the unregulated input is 10.0 V. What should be the value of series resistor R_L?**

Sol. For a good load regulation, choose $I_z = 5I_L = 20mA$ then, $I_s = I_z + I_L = 24$ mA
$\because \quad V_s = 10 - V_z = 4V$

$$\therefore \quad R_s = \frac{V_s}{I_s} = \frac{4V}{24\,mA} = 167\,\Omega$$

The nearest value of carbon resistor is 150 Ω. So, a series resistor of 150 Ω is appropriate.

Q. **Determine the range of values of V_i that will maintain the Zener diode in ON state.**



Sol. Since $I_L = \frac{20}{1.2\,k\Omega} = 16.67\,mA.$

To maintain the zener diode ON

$I_{R_{min}} = I_L$

$\therefore \quad V_{i_{min}} = V_z + I_{R_{min}} R$

$= 20 + 16.67 \times 10^{-3} \times 220 = 23.67\ V$

$I_{R_{max}} = I_L + I_{Zm}$

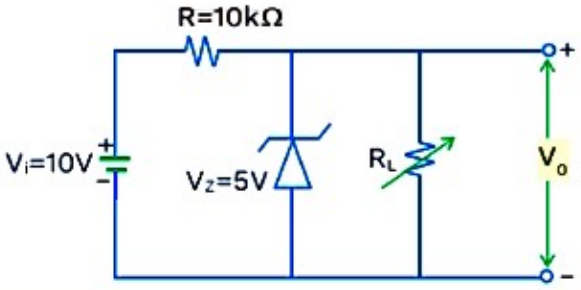
$= (16.67 + 60)\ mA$

$\therefore \quad V_{i_{max}} = V_Z + RI_{R_{max}}$

$= 20 + 220 \times 76.67 \times 10^{-3}$

$= 20 + 16.87 = 36.87\ V$

Q. **In Zener diode circuit if R_L is varied from 1 kΩ to 100 kΩ. Derive output voltage V_0 as a function of the resistance value.**



Sol. Given, R_L is varied from 1kΩ to 100 kΩ.

Since $V_0 = \frac{R_L V_i}{R_L + R}$ .....(i)

For $R_L < 10\ k\Omega$, $V_0 < 5\ V$, (i.e. $V_0 < V_Z$)

Ideal Zener diode behaves like open circuit in this condition and output voltage is related to load resistance by equation (i)

From the above equation the value of R_L that will turn the Zener diode ON is

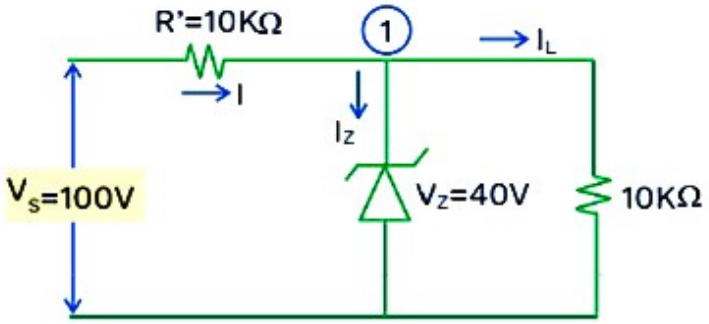
$$R_{L_{min}} = \frac{RV_z}{V_i - V_z} = \frac{(10k\Omega)(5)}{(10-5)} = 10k\Omega$$

Here $V_0 = V_z = 5V$

After turn on, Zener diode maintains the output voltage equal to V_z, therefore, V_0 will be independent of R_L and output voltage will be constant and equal to V_z

$\Rightarrow \quad V_0 = V_z = 5V$

Q.



A voltage regulator circuit is shown in above figure, calculate:
(a) The output voltage.
(b) Voltage drop across series resistance.
(c) The current through Zener diode.

Sol. (a) Zener diode has a property that it maintains voltage across it constant (equal to V_z), so output voltage = V_Z = 40V

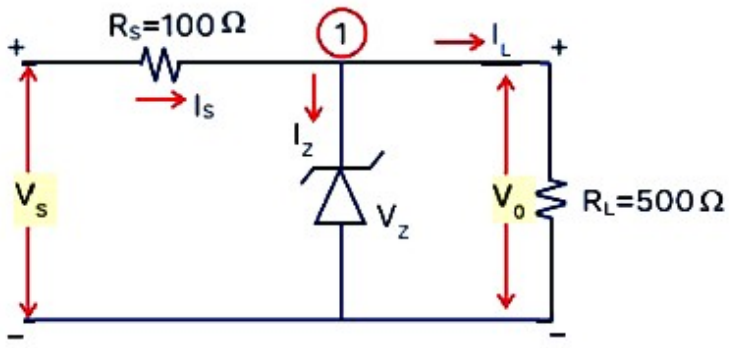
(b) Voltage drop across series resistance R' = $V_s - V_Z$ = 100 – 40 = 60V

(c) So current, $I = \frac{60}{10 \times 10^3} = 6\text{mA}$

Load current $I_L = \frac{40}{10 \times 10^3} = 4\text{mA}$

Applying KCL at point (1); $I_Z = I - I_L = 6 - 4 = 2\text{mA}$

Q.



Find the range of input voltage V_s in figure, for which Zener diode maintain 50V across output. Given that Zener current rating is 50 mA.

Sol. For minimum input voltage, $I_z = 0$

From KCL at node (1), $I_s = I_z + I_L$

So for minimum input voltage $I_s = I_L$

$$I_L = \frac{V_0}{500} = \frac{50}{500} = 0.1A = 100\,mA \Rightarrow I_s = 0.1A$$

Minimum input voltage $= V_0 + I_s R_s$

$= 50 + I_s R_s$

$= 50 + 0.1 \times 100 = 60V$

For maximum input voltage, rated current will flow through Zener diode so

For maximum input voltage, $I_z = 50$ mA

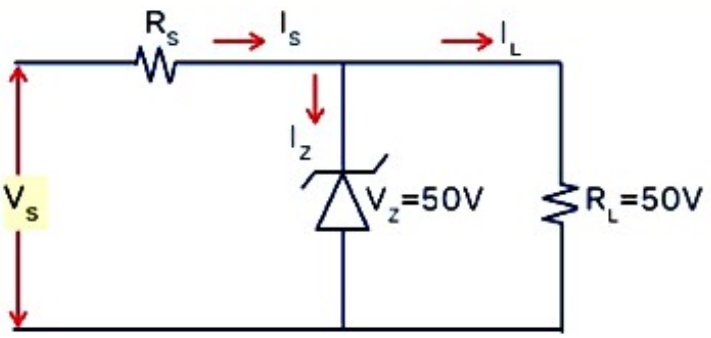
$I_s = I_L + I_z = 100 + 50 = 150$ mA

Maximum input voltage $= V_0 + I_s R_s = 50 + I_s R_s$

$= 50 + 150 \times 10^{-3} \times 100$

$= 50 + 15 = 65V$

Q.



A Zener diode is used to regulate the output voltage at 50 V as shown in figure. Output voltage is maintained constant as long as I_z varies 100 mA to 1A. Calculate resistance 'R_s', if input voltage varies between 60 to 100 V.

Sol. Series resistance 'R_s' is selected according to maximum value of I_s

$$(I_S)_{max} = (I_Z)_{max} + I_L$$

$$I_L = \frac{V_0}{R_L} = \frac{50}{50} = 1A$$

$$(I_{Zmax}) = 1A$$

$$(I_S)_{max} = 1 + 1 = 2A$$

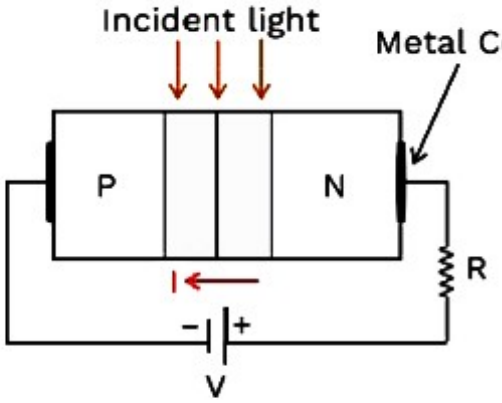
Min. current will flow when input voltage (V_s) is at its minimum value (60V)

$$V = \frac{(V_S)_{max} - V_0}{(I_S)_{max}} = \frac{100 - 50}{2} = \frac{50}{2} = 25\,\Omega$$

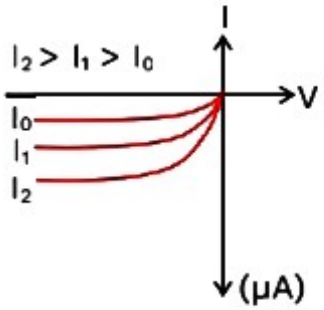
Optoelectronic Junction Devices

Those semiconductor diodes in which carriers are generated by photons (photo-excitation), are called optoelectronic devices.

1. Photo-diode :



An illuminated photo diode



I-V characteristics

The region of operation of photo-diode is limited to only reverse bias. When there is no illumination, a reverse saturation current or dark current (I_0) flows due to thermally generated minority carriers. This current is independent of applied reverse voltage.

When light falls at the P-N junction, light photons impart their energy to valence electrons to break the covalent bond. So electron and hole pairs are generated and their concentration depends upon energy of photons (i.e. intensity of light). These additionally generated electron-hole pairs are added to already present minority carriers and total quantity of minority carriers increases. Thus total reverse current (I) increases. Thus,

$$I = I_0 + I_z$$

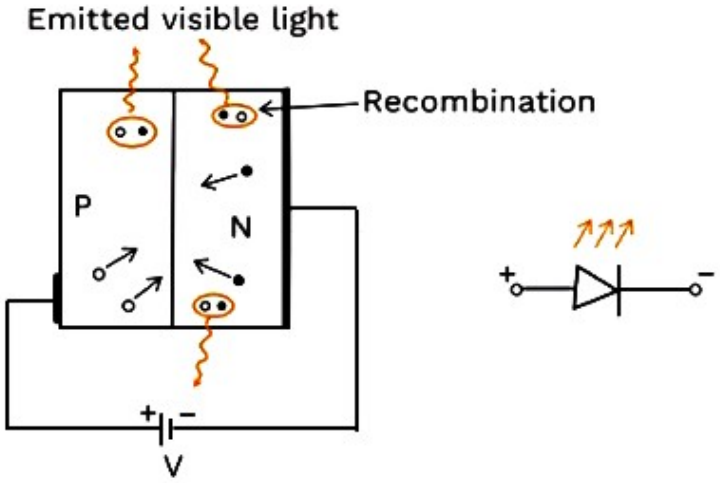
where I_z = current due to photo generated minority carriers,

I_0 = current due to thermally generated minority carriers or dark current

Applications :

(i) Light detection system.
(ii) Counting of objects.
(iii) Reading of film sound track.
(iv) Light operating switches.
(v) High speed reading of computer punched cards and tapes.
(vi) Fire alarm system.

2. Light Emitting Diode (LED) :



Light emitting diode and its symbol

LED works in a forward bias condition. Metal contacts are made both on P- and N-side for external connection to battery. The contact surface to P-side is much smaller, to permit the emergence of maximum light. LED is a direct band gap type diode.

When electron is excited from valence band to conduction band, the electrons and hole pairs are created. Electrons stay in conduction band for a very short period called life time ($\simeq 10^{-9}$ sec) and then return back. In transition from higher energy state (conduction band, E_c) to lower energy state (valence band, E_v), these electrons release energy.

Energy released, $E_g = E_c - E_v$

and associated frequency of radiation, $f = \frac{E_g}{h}$

where h = plank's constant.

For Si and Ge diode, emitted radiation falls in the infrared region. LEDs may be made from semiconducting compounds like gallium arsenide or indium phosphide.

Application :

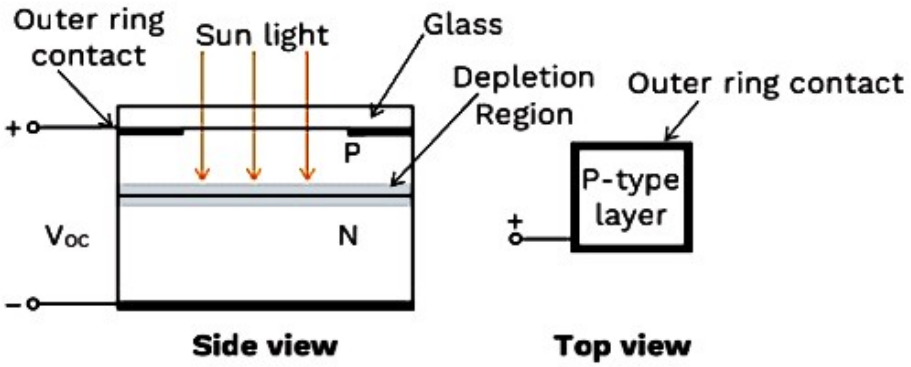
(i) LED is designed so as to produce coherent light with a very narrow band width. We call this LASER diode and find application in optical communication system and in CD player.

(ii) As an opto-isolator (combination of LED and photodiode in same package).

(iii) In display devices.

3. Solar Cell :

It converts solar energy into electrical energy. Although efficiency of conversion is very poor (10% – 40%).



Solar cell consists of a P-N junction with metallic contact at bottom and metallic outer ring at top along with glass. Every effort is made to ensure that the surface area perpendicular to the sun light ray is maximum. Thickness of P-type material is kept smaller so that light photons reach at P-N junction easily. Se, Si, GaAs, InAs, CdS are widely used materials.

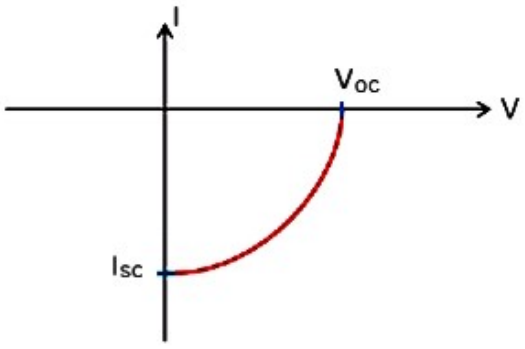
Operation of solar cell is similar to a photo diode. Light photons fall on the junction and impart their energy to valence electrons and dislodge them. Consequently, a large number of free electrons and holes are created on each side of junction. Holes generated in N-side and electrons generated in P-side material move towards P and N region, respectively. The depletion layer does not oppose the flow of minority carriers. Therefore, a minority current flows which in turn depends upon light intensity.

If terminals are shorted, a current flows, which is known as short circuit current (I_{sc}). The short circuit current without any sun light falling on the junction is called dark current.

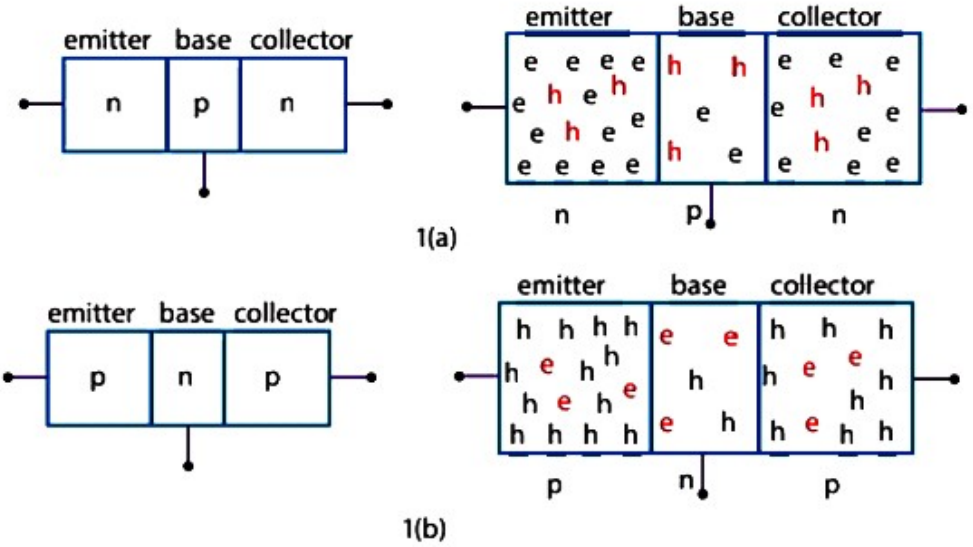
Under open circuit condition, a voltage is obtained which is called photo voltaic e.m.f (V_{oc}) and its magnitude is around 0.4 V.

For getting higher voltage, a string of series connected solar cells is used. For increasing current rating, string of parallel connected solar cells is used.

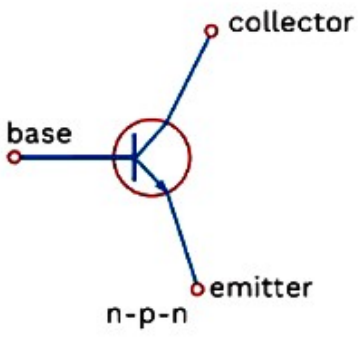
I-V Characteristics :



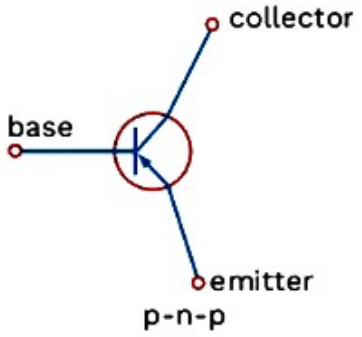
Junction Transistors



A junction transistor is formed by sandwiching a thin layer of a p-type semiconductor between two layers of n-type semiconductors or by sandwiching a thin layer of an n-type semiconductor between two layers of p-type semiconductors. In figure 1(a), we show a transistor in which a thin layer of a p-type semiconductor is sandwiched between two n-type semiconductors. The resulting structure is called an n-p-n transistor. In figure 1(b), we show a p-n-p transistor, where an n-type thin layer is sandwiched between two p-type layers. In actual design, the middle layer is very thin (≈ 1 μm) as compared to the widths of the two layers at the sides. The middle layer is called the base and is very lightly doped with impurity. One of the outer layer is heavily doped and is called emitter. The other outer layer is moderately doped and is called collector. Usually, the emitter-base contact area is smaller than the collector-base contact area. Terminals come out from the emitter, the base and the collector for external connections. Thus, a transistor is a three-terminal device.



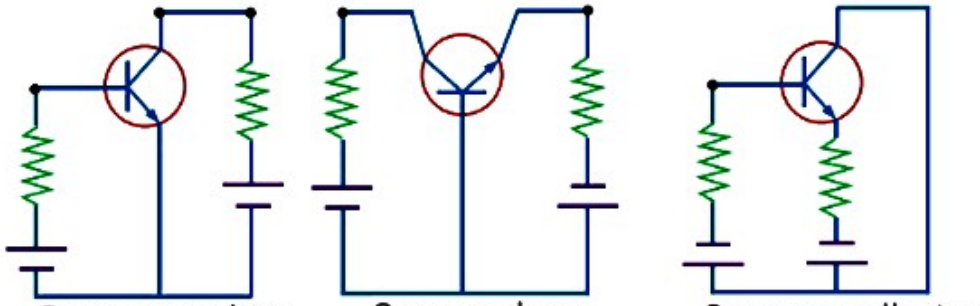
2(a)



2(b)

Figure (2) shows the symbols used for a junction transistor. In normal operation of a transistor, the emitter-base junction is always forward-biased whereas the collector-base junction is reverse-biased. The arrow on the emitter line shows the direction of the current through the emitter-base junction. In an n-p-n transistor, there are a large number of conduction electrons in the emitter and a large number of holes in the base. If the junction is forward-biased, the electrons will diffuse from the emitter to the base and holes will diffuse from the base to the emitter. The direction of electric current at this junction is, therefore, from the base to the emitter. This is indicated by the outward arrow on the emitter line in figure 2(a). Similarly, for a p-n-p transistor the current is from the emitter to the base when this junction is forward biased which is indicated by the inward arrow in figure 2(b).

Biasing

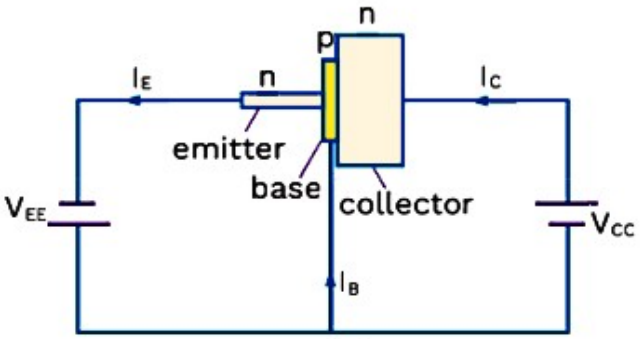


Suitable potential differences should be applied across the two junctions to operate the transistor. This is called biasing the transistor. A transistor can be operated in three different modes: common-emitter (or grounded-emitter), common-collector (or grounded-collector) and common-base (or grounded-base). In common-emitter mode, the emitter is kept at zero potential and the other two terminals are given appropriate potentials (figure 3(a)). Similarly, in common-base mode, the base is kept at zero potential (figure 3(b)) whereas in common-collector mode, the collector is kept at zero potential (figure 3(c)).

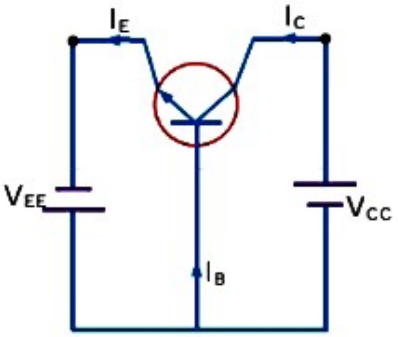
Working of a Transistor

Let us consider an n-p-n transistor connected to the proper biasing batteries as shown in figure below. In part 4(a) of the figure, a physical picture of the transistor is used whereas in part 3(b), its symbol is used. Let us look at the current due to electrons. The emitter-base junction is forward-biased, so electrons are injected by the emitter into the base. The thickness of the base region is so small that most of the electrons diffusing into the base region cross over into the collector region. The reverse bias at the base-collector junction helps this process, because, as the electrons appear near this junction, they are attracted

by the collector. These electrons go through the batteries V_{CC} and V_{EE} and are then back to the emitter.



4(a)



4(b)

The electrons going from the battery V_{EE} to the emitter constitute the electric current I_E in the opposite direction. This is known as emitter current. Similarly, the electrons going from the collector to the battery V_{CC} constitute the collector current I_C.

We have considered only the current due to the electrons. Similar is the story of the holes which move in the opposite direction but result in current in the same direction. Currents I_E and I_C refer to the net currents. However, since the base is only lightly doped, the hole concentration is very low and the current in an n-p-n transistor is mostly due to the electrons. As almost all the electrons injected into the emitter go through the collector, the collector current I_C is almost equal to the emitter current. In fact, I_C is slightly smaller than I_E because some of the electrons coming to the base from the emitter may find a path directly from the base to the battery V_{EE}. This constitutes a base current I_B. The physical design of the transistor ensures that such events are small and hence I_B is small. Typically, I_B may be 1% to 5% of I_E.

Using Kirchhoff's law, we can write,

$$I_E = I_B + I_C \qquad \ldots(1)$$

α and β parameters:

α and β parameters of a transistor are defined as

$$\alpha = \frac{I_C}{I_E} \text{ and } \beta = \frac{I_C}{I_B} \qquad \ldots(2)$$

Using equation (1)

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

or $$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

or $\beta = \frac{\alpha}{1-\alpha}$ ...(3)

As I_B is about 1–5% of I_E, α is about 0.95 to 0.99 and β is about 20 to 100.

Transistor Used in an Amplifier Circuit

Figure (5) shows an amplifier circuit using an n-p-n transistor in common-emitter mode. The battery E_B provides the biasing voltage V_{BE} for the base-emitter junction. A potential difference V_{CE} is maintained between the collector and the emitter by the battery E_C. The base-emitter junction is forward biased and so the electrons of the emitter flow towards the base. As the base region is very thin - of the order of a micrometre - and the collector is also maintained at a positive potential, most of the electrons cross the base region and move into the collector. As discussed earlier, the current I_C is about 0.95 I_E to 0.99 I_E.

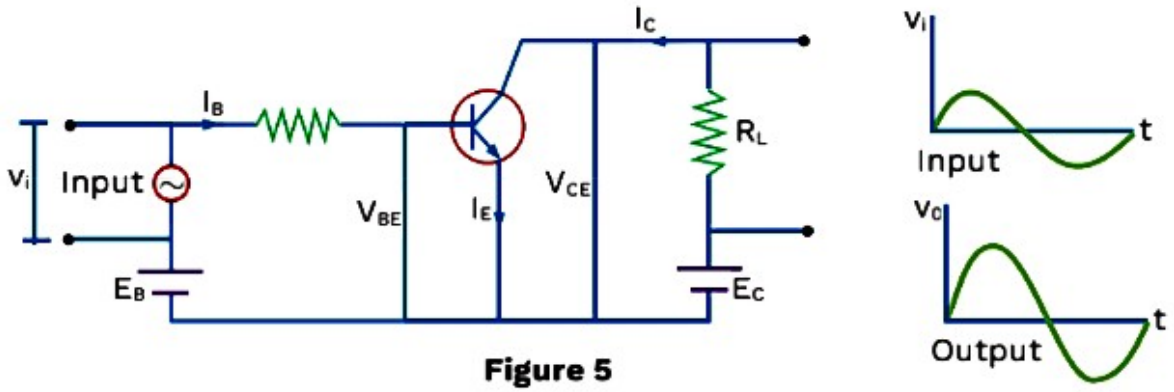


Figure 5

The holes in the base region may diffuse into the emitter due to the forward biasing of the base-emitter junction. Also, the electrons coming from the emitter may recombine with some of the holes in the base. If the holes are lost in this way, the base will become negatively charged and will obstruct the incoming electrons from the emitter. If the base current I_B is increased by a small amount, the effect of hole-diffusion and hole-electron recombination may be neutralised, and the collector current will be increased. Thus, a small change in the current I_B in the base circuit controls the larger current I_C in the collector circuit. This is the basis of amplification with the help of a transistor.

The input signal, to be amplified, is connected in series with the biasing battery E_B in the base circuit. A load resistor having a large resistance R_L is connected in the collector circuit and the output voltage is taken across this resistor. As the potential difference V_{BE} changes with time due to the input signal, the base current I_B changes. This results in a change ΔI_C in the collector current. The current gain, defined as $\Delta I_C/\Delta I_B$, is typically of the order of 50. The change in the voltage across R_L is, accordingly,

$$\Delta V = R_L \Delta I_C$$

Thus, an amplified output is obtained across R_L.

Voltage gain, Current gain and Power gain

When a signal voltage υ_i is added in the base circuit, the voltage across the load resistance changes by υ_0. The ratio $\frac{\upsilon_0}{\upsilon_i}$ is called the voltage gain of the amplifier.

Suppose the input signal has a voltage υ_i at an instant. This produces a change in the base current I_B. As the base-emitter junction is forward biased, it offers a small dynamic resistance R_{BE}. The change in the current in the base circuit is

$$\Delta I_B = \frac{\upsilon_i}{R_{BE}}$$

The resistance R_{BE} is also called the input resistance of the circuit.
The collector current I_C is related to I_B as

$$I_C = \beta I_B$$

Thus, the change in current I_C due to the signal voltage is,

$$\Delta I_C = \beta \Delta I_B = \beta \frac{\upsilon_i}{R_{BE}}$$

The output voltage, i.e., the change in the voltage across the load resistance is

$$\upsilon_0 = \Delta V = R_L \Delta I_C = \frac{\beta \upsilon_i R_L}{R_{BE}}$$

The voltage gain is

$$\frac{\upsilon_0}{\upsilon_i} = \beta \frac{R_L}{R_{BE}}$$

As β is of the order of 50 and R_L may be much larger than R_{BE}, the voltage gain is high.
As mentioned earlier, the current gain is defined as the change in the collector current divided by the change in the base current when the signal is added in the base circuit.
Thus, the current gain is

$$\frac{\Delta I_C}{\Delta I_B} = \beta$$

Power gain = voltage gain × current gain

$$= \frac{\beta^2 R_L}{R_{BE}}$$

Transfer Conductance

To have a large amplification, a small change in V_{BE} should result in a large change in the collector current I_C. This property is measured by a quantity transfer conductance g_m defined as

$$g_m = \frac{\Delta I_C}{\Delta V_{BE}}$$

It is also known as transconductance.

Transistor Used in an Oscillator Circuit

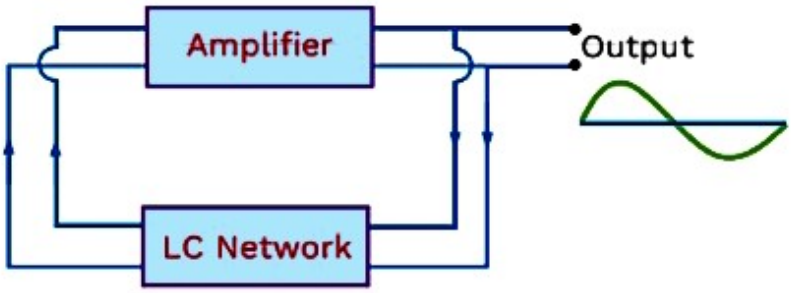


Figure 6

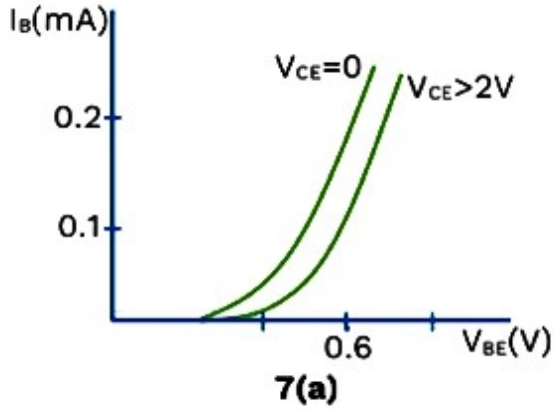
The function of an oscillator circuit is to produce an alternating voltage of desired frequency when only DC batteries are available. Figure (6) shows a schematic representation of an oscillator circuit. The basic parts in this circuit are (a) an amplifier and (b) an LC network. The amplifier section is just a transistor used in common-emitter mode. The LC network consists of an inductor and a capacitance. This network resonates at a frequency

$$\nu_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$$

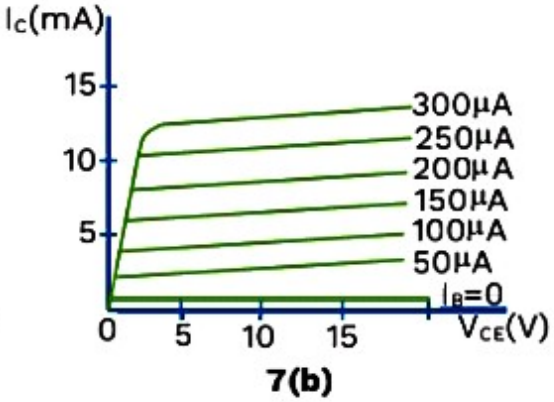
Batteries are used to bias the transistor and no external input signal is fed to the amplifying section. A part of the output signal is fed back to the input section after going through the LC network. This signal is amplified by the transistor and a part is again fed back to its input section. Thus, it is a self-sustaining device. The component with the proper frequency ν_0 gets resonantly amplified and the output acts as a source of alternating voltage of that frequency. The frequency can be varied by varying L or C.

Transistor Characteristics

Let us consider an n-p-n transistor in common-emitter configuration as in figure (5). We can view this circuit as made of an input section and an output section. The input section contains the base-emitter junction and the voltage source whereas the output section contains the base-collector junction and the voltage source. The current I_B may then be called the input current and the current I_C the output current. The voltage applied to the base emitter junction, i.e., in the input section is V_{BE} and that applied to the base-collector junction, i.e., in the output section, is V_{CE}. When the input current I_B is plotted against the voltage V_{BE} between the base and the emitter, we get the input characteristics. Similarly, when the output current I_C is plotted against the voltage V_{CE}, we get the output characteristics.



7(a)



7(b)

These characteristics are shown in figure (7). The input characteristic shown in figure 7(a) are like those of a forward-biased p-n junction. If the biasing voltage is small as compared to the height of the potential barrier at the junction, the current I_B is very small. Once the voltage is more than the barrier height, the current rapidly increases. However, since most of the electrons diffused across the junction go to the collector, the net base current is very small (in microamperes) even at large values of V_{BE}.

The output characteristics are shown in figure 7(b). For small values of the collector voltage, the collector-base junction is reverse-biased because the base is at a more positive potential. The current I_C is then small. As the electrons are forced from the emitter side, the current I_C is still quite large as compared to a signal reverse biased p-n junction. As the voltage V_C is increased, the current rapidly increases and becomes roughly constant once the junction is forward-biased. For higher base currents, the collector current is also high and increases more rapidly, even in forward bias.

Q. **A change of 8.0 mA in the emitter current brings a change of 7.9 mA in the collector current. How much change in the base current is required to have the same change 7.9 mA in the collector current? Find the values of α and β.**

Sol. We have,

$$I_E = I_B + I_C$$

or $\Delta I_E = \Delta I_B + \Delta I_C$

From the question, when ΔI_E = 8.0 mA, ΔI_C = 7.9 mA.

Thus,

$$\Delta I_B = 8.0 \text{ mA} - 7.9 \text{ mA} = 0.1 \text{ mA}.$$

So, a change of 0.1 mA in the base current is required to have a change of 7.9 mA in the collector current.

$$\alpha = \frac{I_C}{I_E} = \frac{\Delta I_C}{\Delta I_E} = \frac{7.9\,mA}{8.0\,mA} \approx 0.99$$

$$\beta = \frac{I_C}{I_B} = \frac{\Delta I_C}{\Delta I_B} = \frac{7.9\,mA}{0.1mA} = 79.$$

Check if these values of α and β satisfy the equation.

$$\beta = \frac{\alpha}{1-\alpha}.$$

Q. **A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 20 mV is added to the base-emitter voltage, the base current changes by 20 μA and the collector current changes by 2 mA. The load resistance is 5 kΩ. Calculate (a) the factor β, (b) the input resistance R_{BE}, (c) the transconductance and (d) the voltage gain.**

Sol. (a) $\beta = \frac{\Delta I_C}{\Delta I_B} = \frac{2\,mA}{20\,\mu A} = 100.$

(b) The input resistance $R_{BE} = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{20\,mV}{20\,\mu A} = 1\,k\Omega$

(c) Transconductance $= \frac{\Delta I_C}{\Delta V_{BE}} = \frac{2\,mA}{20\,mV} = 0.1$ mho.

(d) The change in output voltage is $R_L \Delta I_C$ = (5 kΩ) (2 mA) = 10V.
The applied signal voltage = 20 mV.

Thus, the voltage gain is = $\frac{10\,V}{20\,mV} = 500.$

Logic Gates

Introduction :

A logic gate is a digital circuit which is based on certain logical relationship between the input and the output voltages of the circuit.

The logic gates are built using the semiconductor diodes and transistors.

Each logic gate is represented by a characteristic symbol.

The operation of a logic gate is indicated in a table, known as truth table. This table contains all possible combination of inputs and the corresponding outputs.

A logic gate is also represented by Boolean algebraic expression. Boolean algebra is a method of writing equations showing how an output depends upon the combination of inputs. Boolean algebra was invented by George Boolean.

Basic Logic Gates :

(1) OR gate (2) AND gate, and (3) NOT gate

(1) The OR gate :- The output of an OR gate attains the state 1 if one or more inputs attain the state 1.

Logic symbol of OR gate:



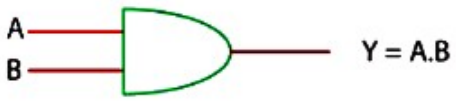
The Boolean expression for OR gate is Y = A + B, read as Y equal A OR B.

Truth table of a two input OR gate:

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(2) The AND gate : The output of an AND gate attains the state 1 if and only if all the inputs are in state 1.

Logic symbol of AND gate:



The Boolean expression of AND gate is Y = A.B It is read as Y equal A AND B .

Truth table of a two input AND gate:

Input		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(3) The NOT gate : The output of a NOT gate attains the state 1 if and only if the input does not attain the state 1.

Logic symbol of NOT gate:



The Boolean expression is $Y = \overline{A}$, read as Y equals NOT A.

Truth table of NOT gate:

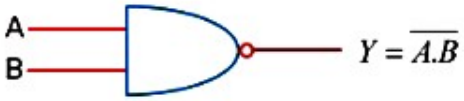
Input	Output
A	B
0	1
1	0

Combination of Gates :

The three basic gates (OR, AND and NOT) when connected in various combinations give us logic gates such as NAND and NOR gates, which are the universal building blocks of digital circuits. Repeated use of NAND gates (or NOR gates) give other gates. Therefore, any digital system can be achieved entirely from NAND or NOR gates.

The NAND gate:

Logic symbol of NAND gate:



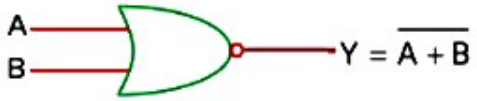
The Boolean expression of NAND gate is $Y = \overline{A.B}$

Truth table of a NAND gate:

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

The NOR gate :

Logic symbol of NOR gate:



The Boolean expression of NOR gate is $Y = \overline{A + B}$

Truth table of a NOR gate:

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Xor and Xnor Gate :

Exclusive – OR gate (XOR gate) : The output of a two–input XOR gate attains the state 1 if one and only one input attains the state 1.

Logic symbol of XOR gate:

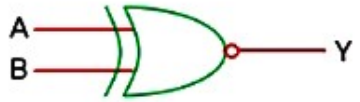


The Boolean expression of XOR gate is $Y = A\bar{B} + \bar{A}B$ or $Y = A \oplus B$

Truth table of a XOR gate:

Input		Output
A	B	A
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive – NOR gate (XNOR gate) : The output is in state 1 when its both input are the same that is, both 0 or both 1.
Logic symbol of XNOR gate:



The Boolean expression of XNOR gate is $Y = A.B + \overline{A}.\overline{B}$ or $Y = \overline{A \oplus B}$

Truth table of a XNOR gate:

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Laws of Boolean Algebra
Basic, OR, AND, and NOT operations are given below

OR	AND	NOT
A + 0 = A	A. 0 = 0	$A + \overline{A} = 1$
A + 1 = 1	A. 1 = A	$A. \overline{A} = 0$
A + A = A	A. A = A	$\overline{\overline{A}} . A = A$

Boolean algebra obeys commutative, associative and distributive laws as given below.
Commutative laws :
A + B = B + A ;
AB = BA

Associative laws :

$A + (B + C) = (A + B) + C$;

$A.(B.C) = (A.B).C$

Distributive laws :

$A(B + C) = AB + AC$

Some other useful identities :

(i) $A + AB = A$

(ii) $A.(A + B) = A$

(iii) $A + \bar{A}B = A + B$

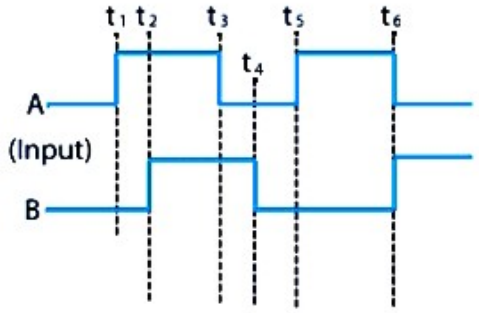
(iv) $A.(\bar{A} + B) = AB$

(v) $A + BC = (A + B)(A + C)$

De Morgan's theorem :

$\overline{A + B} = \bar{A}.\bar{B}$ and $\overline{A.B} = \bar{A} + \bar{B}$

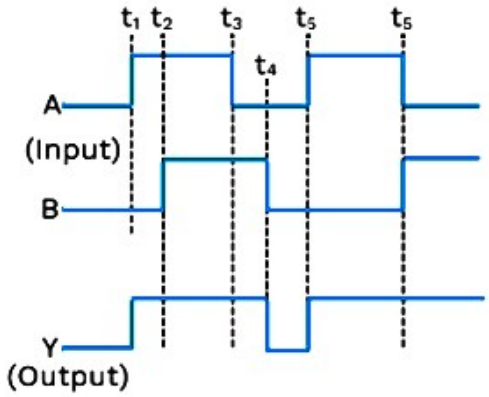
Q. **Justify the output waveform (Y) of the OR gate for the following inputs A and B given in figure below.**



Sol. Note the following :

- At $t < t_1$; $A = 0, B = 0$; Hence $Y = 0$
- For t_1 to t_2; $A = 1, B = 0$; Hence $Y = 1$
- For t_2 to t_3; $A = 1, B = 1$; Hence $Y = 1$
- For t_3 to t_4; $A = 0, B = 1$; Hence $Y = 1$
- For t_4 to t_5; $A = 0, B = 0$; Hence $Y = 0$
- For t_5 to t_6; $A = 1, B = 0$; Hence $Y = 1$
- For $t > t_6$; $A = 0, B = 1$; Hence $Y = 1$

Therefore, the waveform Y will be as shown in the Figure.

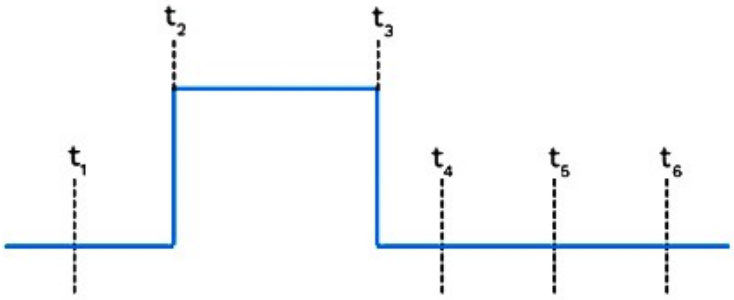


Q. **Take A and B input waveforms similar to that in previous example. Sketch the output waveform obtained from AND gate.**

Sol.

- For $t \leq t_1$; A = 0, B = 0; Hence Y = 0
- For t_1 to t_2; A = 1, B = 0; Hence Y = 0
- For t_2 to t_3; A = 1, B = 1; Hence Y = 1
- For t_3 to t_4; A = 0, B = 1; Hence Y = 0
- For t_4 to t_5; A = 0, B = 0; Hence Y = 0
- For t_5 to t_6; A = 1, B = 0; Hence Y = 0
- For $t > t_6$; A = 0, B = 1; Hence Y = 0

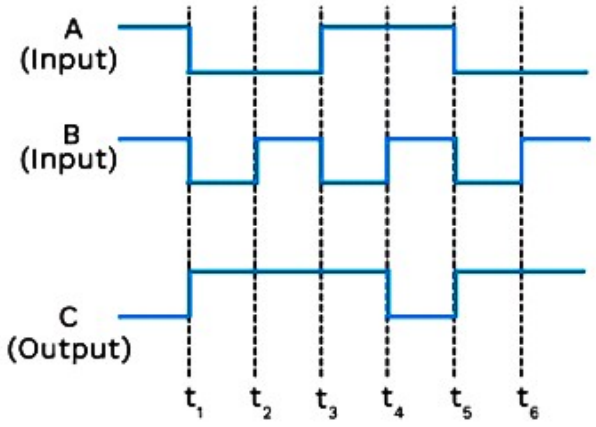
Based on the above, the output waveform for AND gate can be drawn as given below.



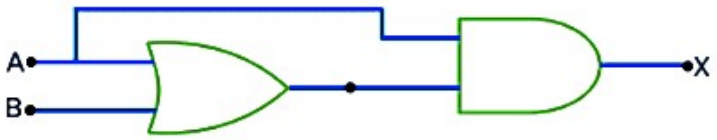
Q. **Sketch the output Y from a NAND gate having inputs A and B given below:**

Sol.

- For $t < t_1$; A = 1, B = 1; Hence Y = 0
- For t_1 to t_2; A = 0, B = 0; Hence Y = 1
- For t_2 to t_3; A = 0, B = 1; Hence Y = 1
- For t_3 to t_4; A = 1, B = 0; Hence Y = 1
- For t_4 to t_5; A = 1, B = 1; Hence Y = 0
- For t_5 to t_6; A = 0, B = 0; Hence Y = 1
- For $t > t_6$; A = 0, B = 1; Hence Y = 1



Q. **Construct the truth table for the function X of A and B represented by figure below.**



Sol. Here an AND gate and an OR gate are used. Let the output of the OR gate be Y. Clearly, Y = A + B. The AND gate receives A and A + B as input. The output of this gate is X. So X = A(A + B). The following table evaluates X for all combinations of A and B. The last three columns give the truth table.

A	B	Y = A + B	X = A(A + B)	A	B	X
0	0	0	0	0	0	0
0	1	1	0	0	1	0
1	0	1	1	1	0	1
1	1	1	1	1	1	1